37. How is the trend of urbanization in Ethiopia explained? Ethiopia has
A. had a slow rate of urbanization in recent years.
B. had a low rate of urbanization after the Italian occupation.
C. a low rate and high level or urbanization in Africa.
D. one of the least level of urbanization in the world.

Explanation: Ethiopia has one of the least level of urbanization (16.2%) and one of the least rate of urbanization in the world.

Answer: D

38. Which one of the following describes economic growth and development trend in Ethiopia?
A. Since 1991 the Ethiopian economy has performed better than it did in the earlier years.
B. There was appreciable economic growth and development during the Dergue regime.
C. Access to quality social services has been decreasing from time to time.
D. Agriculture sector still remains the largest contributor of the Gross Domestic Product (GDP).

Explanation: Agricultural sector is still the largest contributor for the Gross Domestic Product of Ethiopia.

Answer: D

39. Among the following service sub-sectors in Ethiopia, which one is the lowest in terms of growth performance over the past two decades?
A. Health
B. Education
C. Public administration and Finance
D. Banking and insurance

Explanation: Public administration and finance show the lowest in terms of growth performance over the past two decades.

Answer: C

40. How is the present feature of Ethiopian socio-economic development described?
A. Ethiopian has a well-functioning manufacturing sector contribut-

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ing lot to socio-economic development.

B. Industrial sector is the leading in terms of contribution to socio-economic development.

C. The socio-economic development of the country is still low compared to that of other countries.

D. Ethiopia has a mature and diversified economy with a high proportion of skilled labor force.

Explanation: The socio-economic development of Ethiopia is still low compared to that of other countries.

Answer: C

41. Which of the following describes the scope of geography?

A. The scope of geography lies beyond the upper atmosphere and the ocean current.

B. Geography is directly interested in investigating thermodynamic characteristics of the interior of the earth's core.

C. Geography focuses more on the outer space than the surface of the earth.

D. Geography focuses on spatial aspects of phenomena between the lithosphere and the lower layer of atmosphere.

Explanation: The scope of geography focused on spatial aspects of phenomena between the lithosphere and the lower layer of the atmosphere.

Answer: D

42. Geographers use systematic approach under which of the circumstances listed below?

A. Exploring the physical and human geography of Africa.

B. Studying every details of the world in a holistic way.

C. Investigation the scale of climate impact on pastoral communications.

D. Describing the general characteristics of the Horn of Africa.

Explanation: Geographers use systematic approach to study the distribution of a single feature over the surface of the earth. For instance, the scale of climate impact on pastoral communities.

Answer: C

43. How would you describe the school of possibilism?

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A. It thinks that human beings can modify nature using technology.
B. It assumes that man is under the mercy of nature.
C. It views that neither environment nor human beings have the upper hand.
D. It is a philosophical thought that gives more emphasis to nature.

Explanation: Environmental possibilism advocates that human beings can modify nature using technology.

Answer: A

44. Which one of the following could be among the conditions that initiated the emergence of applied geography?
A. The urge to answer the where and what questions across the world
B. The growing need to solve socio-economic and political problems
C. The emergence of primitive communities
D. Discovery of the unknown part of the world

Explanation: The growing need to solve socio-economic and political problems is one of the conditions that initiated the emergence of applied geography.

Answer: B

45. Among the following disciplines, which one has a more similar concern with population geography?
A. Meteorology
B. Sociology
C. Demography
D. Economics

Explanation: Demography has more similar concern with population Geography. It is the study of human population, including their size, growth, density, etc.

Answer: C

46. What type of drainage pattern develops in an area where the tributaries join the main river at right angles?
A. Trellis B. Dendritic C. Centripetal D. Radial

Explanation: In trellis drainage pattern tributaries join the main river nearly at right angle.

Answer: A

47. In which of the following conditions does a river capture occur?

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A. When a victim stream erodes the course of a pirate stream
B. When a pirate stream overrides a victim stream
C. When the rock on the direction of a pirate stream is softer
D. When heavy rain showers the watershed of a victim stream

Explanation: River capture occur when the rock on the side of the victim river is softer, heavy rainfall on the pirate's river side of the watershed, the pirate river has steeper course and erode it and the watershed consists of lower round.

Answer: B

48. Which one of the following features describes the middle course of a river?

A. Oxbow lakes
B. Waterfalls
C. V-shaped valleys
D. Meanders

Explanation: In the middle course the steepness of the river channel decreases. The river began to meander especially in areas adjacent to the lower course.

Answer: D

49. Assume that you are traveling to the southern region of Ethiopia, crossing Hawassa town on the way to the Gedeo Zone. During your travel, what type of settlement pattern would you be more likely to observe?

A. Nucleated B. Dispersed C. Ribbon D. Linear

Explanation: Linear (ribbon) settlement has elongated shape. It develop along road route or ridge.

Answer: C

50. Among the following, which one is responsible for the establishment of Wereta Town at its present location?

A. Hot spring
B. Fogera basin
C. Mineral resources
D. Tis Isat fall

Explanation: One of the main reason for the establishment of Wereta town is intensively cultivated Fogera basin drained by the Rib and Gumera rivers.

Answer: B

51. Among the following, which one explains about indicators of development in Ethiopia?

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A. There has been a declining literacy rate in country recently.
B. Infant mortality rate is below the average of the Sub-Saharan countries.
C. A greater proportion of rural population has access to electricity.
D. There has been an increasing of access to clean water in recent years.

Explanation: One of the indicator of development in Ethiopia is an increasing access to clean water in recent years.

Answer: D

52. The Great Ethiopian Renaissance Dam will open a new chapter for the socio-economic development of the country. Which of the following could be the contribution of this dam?

A. Providing supply of potable water
B. Promoting water transport of the country
C. Promoting tourism development
D. Regulating weather condition of the country

Explanation: Despite its potential for generating HEP and irrigation, GERD could also contribute for tourism and fishing industries.

Answer: C

53. From which of the following regions, has Ethiopia been importing most items in recent years?

A. Europe and Australia
B. The Middle eat and Asia
C. The Far east and Central America
D. North America and Africa

Explanation: According to ministry of Trade and Industry, Ethiopia's most important source imports in 2004/05 were the middle East and Asia which accounted for 57.5%.

Answer: B

54. Which one of the following was the focus of the Plan for Accelerated and Sustained Development to End Poverty (PASDEP) strategy in Ethiopia?

A. Encouraging rapid population growth
B. Promoting rapid non-public sectors

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C. Encouraging traditional subsistent farming

D. Expanding labor- intensive production activities

Explanation: One of the initative of PASDEP creating is jobs. The goals were managing the dynamics of population growth and expanding labor- intensive production activities.

Answer: D

55. What is the **demerit** of migration to the receiving country?

A. It reduces pressure on resources.

B. It causes a decline in birth rate.

C. It triggers division among families.

D. It creates shortage of labour force.

Explanation: The demerits of migration for receiving country are pressure on job, low quality house, ethnic tention, limited skill worker, language difficulties, difficulty to practic their own religion, culture, etc.

Answer: No answer

56. Which one of the following characterizes urbanization in Africa?

A. Low level and high rate of urbanization

B. High level and low rate of urbanization

C. Low level and low rate of urbanization

D. High level and high rate of urbanization

Explanation: Urbanization in Africa is at lowest level (only 38%). However, rate of urbanization is very fast.

Answer: A

57. One among the following is a major problem that affects urban centres of Africa. Which one is it?

A. Air pollution

B. Squatter settlement

C. Shortage of labour force

D. Nucleated settlement

Explanation: Urban centers in Africa are affected by squatter settlement, urban slum, environmental degradation, poor sanitation, unemployment, poverty, social evils, etc.

Answer: B

58. Recently, the government of Ethiopia has reported the rise of per cap-

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ita income of Ethiopian nationals to 1000 US dollars per year. What does this imply?

A. High income groups are expected to earn 1000 US dollar per year.
B. Low income groups actually earn 1000 US dollar per year.
C. Each Ethiopian citizen receives 1000 US dollar per year from the government.
D. Ethiopian citizens, on average, expect to earn 1000 US dollar per year.

Explanation: As per government report the per capita income of Ethiopians is 1000 USD/year. It imply that, Ethiopian citizens, on average, receives 1000 USD per year.

Answer: D

59. Which of the statements below explains the Africa economy?
A. Most African countries have a negative trade balance.
B. Manufacturing sector is the backbone of the economy.
C. Africa has a diversified economy.
D. Mining is the main economic activity.

Explanation: Most African countries have a negative trade balance. This is resulted from export of raw material and import of expensive manufacturing products.

Answer: A

60. Which one best described the human resources of Africa?
A. Most of the African population is underemployed.
B. Africa has a highly skilled manpower.
C. There is a low working labour force potential in the continent.
D. Africa has effectively utilized its human resources potential.

Explanation: The potential of mineral resource is large and varies. However, the level of utilization of the resource very low.

Answer: A

61. How would you explain the potential and the actual resources exploitation in Africa?
A. There is over utilization of resources in the continent.
B. There is a sustainable use of natural resources.
C. Many of the resources of the continent are untouched.

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D. The continent is potentially poor in natural resources.

Explanation: The potential of mineral resource is large and varied. However, the level of utilization of the resource very low.

Answer: C

62. Which of the statements below describes research?

A. It is a systematic way of inventing new things.

B. It is a modern way of recording historical phenomena.

C. It is a way of exploring geographical areas.

D. It is scientific procedure of investigation problems.

Explanation: Research is a scientific method of investigating answer for already identified problem.

Answer: D

63. When is a quantitative research approach applied in geography?

A. In analyzing data from key informant interviews

B. In verifying a particular geographical theory

C. In describing the culture of a society

D. In content analysis and narrations of an event

Explanation: Quantitative research approach is applied when a given theory is tested and verified.

Answer: B

64. What makes geographic research different from research in other disciplines?

A. It mainly focuses on the spatial distribution of phenomena.

B. It mainly emphasizes the temporal variation of things.

C. It greatly emphasizes the cultural values and norms.

D. It assesses demand and supply of the production function.

Explanation: Geographic research is different from other in that it mainly focuses on the spatial distribution of phenomena.

Answer: A

65. Under which of the following is Dukam Town mainly categorized?

A. Commercial town

B. Mining town

C. Market town

D. Industrial town

Explanation: Dukam Town categorized into industrial town. Such

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towns are engaged in processing raw materials into finished goods.

Answer: D

66. Which one of the following governments initiated the first application of GIS to analyze land inventory data?

A. The Canadian Government
B. The USA Government
C. The Norwegian Government
D. The Australian Government

Explanation: The first GIS application was developed by Canadian government to analyze land inventory data.

Answer: A

67. Suppose you are assigned as a field supervisor to assess crises in locust infested areas of Ethiopia. How does GIS primarily assist you in the assessment of the crisis?

A. To count the locusts in the study area
B. To map hotspots of the study area
C. To predict future occurrence of the crises
D. To broadcast the situation of the crises

Explanation: When we are assigned to assess crises in locusts infected areas of Ethiopia, GIS primarily assists us to map hotspots of the study area.

Answer: B

68. What is the relative location of Africa?

A. It is found west of the Atlantic Ocean.
B. Its extreme north is Cape Verde.
C. It is located south of the Mediterranean Sea.
D. It is located mainly in the temperature zone.

Explanation: Africa is located south of Mediterranean sea, east of Atlantic ocean, NW of Indian ocean and west of Red seas and Gulf of Aden.

Answer: C

69. One of the following is the impact of smooth and straight coastline of Africa. Which one is it?

A. The coastlines remain conducive for maritime trade.
B. Construction of man-made ports is an expensive endeavor.

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C. Significant influence of maritime climate in the interior areas.

D. Ease of sea transport and access to important sea routes.

Explanation: Africa has smooth and straight coast line. This makes the continent poor in natural harbors. As a result, construction of man made parts is a expensive endeavor.

Answer: B

70. Which one of the following describes Northern Africa?

A. It is a region where most highlands of the continent are found.

B. It is the largest in areal coverage among regions of the continent.

C. It is the second most populous region in the continent.

D. It is a region with the most diverse economy in the continent.

Explanation: Northern Africa is the largest in area cover among regions of Africa. It covers about 1/3rd of African's land area.

Answer: B

71. Among the following, which one indicates geographical location of African regions?

A. Western Africa is found to the west of the Atlantic Ocean.

B. The horn of Africa is located in the central part of Africa.

C. The Maghreb region is found in northern Africa.

D. Eastern Africa is located to the northeast of the Indian Ocean.

Explanation: Western Africa is found east of Atlantic ocean. The Horn of Africa is located in the NE part of Africa. The maghreb region is found in the North Africa. Eastern Africa is found in the NW of Indian ocean.

Answer: C

72. Which one of the following geological eras was known for extensive denudation and peneplanation?

A. The Mesozoic Era

B. The Cenozoic Era

C. The Precambrain Era

D. The Paleozoic Era

Explanation: Paleozoic era (from 600-250 million years) were known by massive denudation un peneplanation activity.

Answer: D

73. In which of the following countries is the highest mountain of Africa located?

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A. Botswana B. Cameroon C. Tanzania D. Algeria

Explanation: Africa's highest mountain i.e., m.t. kilimanjaro which is 5895m above sea level is found in Tanzania.

Answer: C

74. In which of the following climatic controls has a major effect on the most parts of eastern Africa?

A. Altitude
B. Latitude
C. Ocean current
D. Distance from the sea

Explanation: Most part of East Africa is a highland. Therefore, the most important climatic control in the region is altitude.

Answer: A

75. Which of the following can be generalized about the characteristics of the major rivers of Ethiopia? The major rivers of Ethiopia?

A. have greater navigational potential.
B. are characterized by gentle slopes.
C. are seasonally fluctuate in their volume.
D. rise from rift valley parts of the country.

Explanation: Rivers of Ethiopia characterized by steep slopes, seasonal fluctuation in their volume, rise from high plateau of the country and frequent waterfalls and rapids.

Answer: C

76. Which of the following explains about the Indian Ocean drainage system in Ethiopia?

A. Its major rives originate from the southeastern highlands.
B. It accounts for the largest annual water flows of the country.
C. The Awash and the Omo-Ghibe are the major rivers of the system.
D. It drains the south western highlands and lowlands.

Explanation: The sources of water Indian drainage system are the southern the southeaster Ethiopia highlands.

Answer: A

77. Among the following, which one is a major significance of Ethiopian rivers?

A. Huge geothermal potential
B. Sources of minerals

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C. Hydro-power generation D. Serve as transport routes

Explanation: Ethiopian rivers have many uses some of the use are source of minerals, hydro-power generation, potable water supply and means transport routes.

Answer: C

78. Which of the following can be the immediate solution fort the current problem that Lake Tana facing?

A. Controlling population pressure on the lake

B. Reducing sediment flow to the lake

C. Preventing deforestation in the areas surroundings lake

D. Avoiding the exotic plant dominating the lake

Explanation: One of the immediate solution for current problem that Lake Tana facing is avoiding the exotic plant dominating the lake.

Answer: D

79. Which factor determine the temporal variation of temperature in Ethiopia?

A. Latitude C. Planetary winds

B. Cloud cover D. Ocean currents

Explanation: One of the reason for temporal variation of temperature in Ethiopia is the distance of the over head sun and its apparent north-south movement across the equator. It is related with latitude.

Answer: A

80. Why does the upper rift valley region of Ethiopia face rain shadow effect? This is mainly because of the effect of

A. latitude. C. mountain barriers.

B. distance from the sea. D. weather systems.

Explanation: Due to its location in the lee wards side of rain bearing equatorial westerly winds, the upper rift valley (Afar triangle) affected by rain shadow.

Answer: C

81. Where is the hottest area of Ethiopia located?

A. Around the lake Chamo

B. In the main Ethiopian rift valley

C. Near Mount Batu

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D. In the Afar triangle

Explanation: The hottest area in Ethiopia is found in the Afar Triangle of the Rift valley.

Answer: D

82. Where are the wettest areas of Ethiopia mainly located?

A. Northwestern highlands
C. Northeastern highlands
B. Southwestern highlands
D. Southeastern highlands

Explanation: The wettest areas of Ethiopia are mainly located in south western highlands.

Answer: B

83. Which of the following places receives the highest amount of rainfall during summer season?

A. Hawassa
B. Samara
C. Shakiso
D. Ambo

Explanation: Summer rainfall region consists of the north west highlands and western lowlands of Ethiopia. Ambo categorize to the northwest highlands. Therefore, it receive summer maximum rain.

Answer: D

84. What is the smallest rainfall region in Ethiopia?

A. Summer rainfall region
B. Winter rainfall region
C. Merged spring, summer and autumn rainfall region
D. Autumn and spring rainfall region

Explanation: Merged spring, summer, and autumn rainfall region is the smallest rainfall region in Ethiopia.

Answer: C

85. Among the following winds, which one dominates the climate of tropical Africa?

A. Westerly winds
C. Polar easterlies
B. Monsoon winds
D. Trade winds

Explanation: Trade wind flows from sub-Tropical High to Equatorial Low dominates the climate of tropical Africa.

Answer: D

86. In which of the following months does Southern Africa experience

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the highest temperature?

A. December, January and February

B. September, October and November

C. June, July and August

D. March, April and May

Explanation: In December, January and February southern part of Africa experience relatively higher temperature.

Answer: A

87. Which of the following countries mostly experiences tropical continental type of climate in Africa?

A. Somalia

B. Tunisia

C. Madagascar

D. Morocco

Explanation: Tropical continental climate occurs north and south of the tropical wet zone, in many parts of western Africa, south Africa and most Madagascar.

Answer: C

88. In which of the following areas of Africa does a highland climate occur?

A. The coastal areas of western Africa

B. The Drakensberg mountain region

C. The Maghreb region

D. The Equatorial rainforest region

Explanation: Highland climate occur in east African Highlands, the Drakensberg mountain region and the Atlas mountain region northern Africa.

Answer: B

89. Where in Africa are most of the drought prone areas located?

A. Among the Mozambique coast

B. In cape Province of South Africa

C. In the Mediterranean region

D. In the Sahel region

Explanation: The Sahel region is found in the southern fringe of the Sahara Desert stretching from Senegal to Somalia. It is the most

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Answer: D

90. To which of the following drainage system does the Congo River belong?

A. The Indian Ocean
B. The Atlantic Ocean
C. The Inland drainage system
D. The Mediterranean Sea

Explanation: Congo River belongs to the Atlantic ocean drainage system.

Answer: B

91. Among the following, which one describes the Mediterranean Sea drainage system in Africa?

A. It has only one major river basin in the continent.
B. It drains the northwestern parts of the continent.
C. It experiences a steady volume of water thought the year.
D. It shares the largest catchment area and annual discharge.

Explanation: Mediterranean sea drainage system is dominated by one major river basin, i.e., Nile basin.

Answer: A

92. Which of the following rivers of Africa form an extensive delta at their mouths?

A. The Awash and Ghibe Rivers
B. The Congo and Orange Rivers
C. The Niger and Zambezi Rivers
D. The Senegal and Zaire Rivers

Explanation: Some African rivers form delatic mouth and mangrove swamps. For instance, the Nile, Niger and Zambezi rivers have extensive delta.

Answer: C

93. The Okovango swamps are found in which of the following countries?

A. Togo
B. Mali
C. Gambia
D. Botswana

Explanation: Okovango swamps found in Botswana.

Answer: D

94. Where in Africa, is the Lake Volta located?

A. Ghana
B. Zimbabwe
C. Egypt
D. Nigeria

Explanation: Lake Volta is an artificial lake stored in Akosombodam

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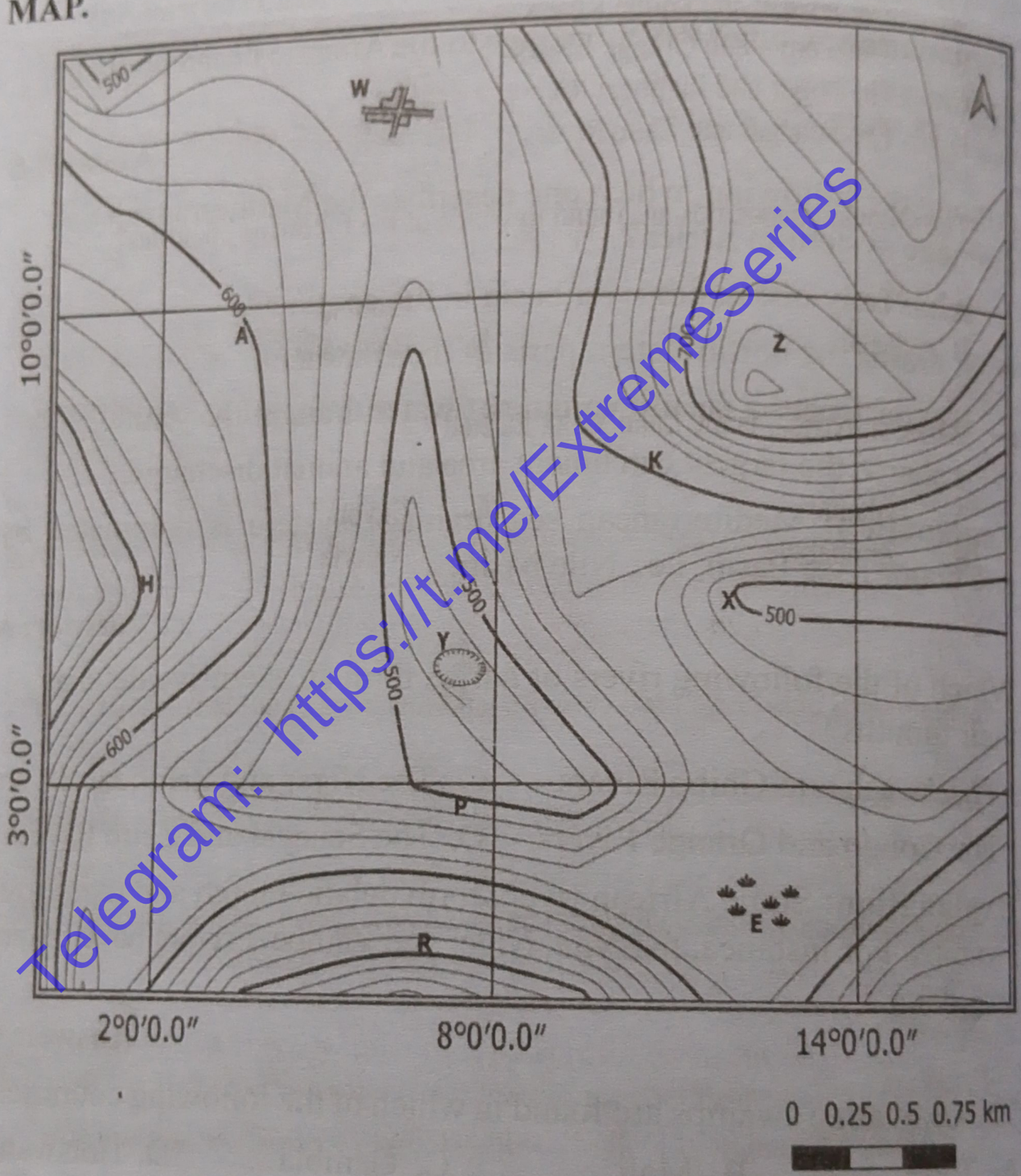
in Ghana.

Answer: A

QUESTIONS 95-100 ARE ABOUT MAP READING. THESE QUESTIONS SHOULD BE DONE ONLY BY NON-BLIND CANDIDATES. QUESTIONS 96, 98 AND 99 REFER TO THE ATTACHED MAP.



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95. Which one describes the difference between map and globe?

A. Globes are visually continent whereas maps are not.

B. A globe is two-dimensional, whereas a map is three-dimensional.

C. Maps are more used than globe in geography.

D. Globes are easy to fold whereas maps are not.

Explanation: Globes are the exact model of the earth. However, maps are more used than globe in geography due to its suitability.

Answer: C

96. What would be the shortest length route for the railway on the contour map between points 'X' and 'H'?

A. 30 km B. 40 km C. 10 km D. 20 km

Explanation:

Given

- Altitude difference between H (700m) and X (500m) is 200m = 0.2km.

- Maximum gradient a train climb is $2\% = \frac{2}{100}$

Required

- The shortest rail route between X and H(MD)

Solution

$$Gr = \frac{AD}{MD} \times 100 \Rightarrow \frac{2}{100} = \frac{0.2km}{MD} \Rightarrow 2MD = 100 \times 0.2km$$

$$\frac{2MD}{2} = \frac{100 \times 0.2km}{2} \Rightarrow MD = 10km$$

Therefore, the shortest possible rail way route between X and H is about 10km.

97. What characterizes zenithal map projection?

A. True distances and directions are reserved.

B. Meridians are concentric circles.

C. Distortion increase from the equator towards pole.

D. Best suited for the projection of tropical regions.

Explanation: Zenithal projection also known as azimuthal projection is retained true distance and direction.

Answer: A

98. What type of geographic feature is represented by the letter 'E'?

A. Forest B. Swamp C. Farmland D. Woodland

Explanation: The geographic feature represented on the attached map by the letter "E" is swamp.

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Answer: A

99. What does the letter 'Y' indicate on the contour map?

A. Spur
B. Cliff
C. Hill
D. Depression

Explanation: Circular contours with tick pointing to the center represent depression. Therefore, letter "Y" represent a depression.

Answer: D

100. Which one of the following distribution maps uses lines to connect points of equal value?

A. Isoplethic maps
B. Diagrammatic maps
C. Chorochromatic maps
D. Choroplethic maps

Explanation: Isoplethic maps are uses lines to connect points of equal value.

Answer: A

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1. In Ethiopian context, which impact of rapid population growth is critical, compared to others?

A. Occurrence of water pollution
C. High rate of deforestation
B. Shortage of pasture land
D. Massive air pollution

Explanation:- Ethiopia high rate of deforestation is caused by rapid population group

Answer: C

2. What important massage do you expect to hear in a speech to be delivered on the population policy of Ethiopia?

A. Discouraging use of contraceptive methods
B. Encouraging early marriage of females
C. Promoting men-focused family planning strategies
D. Establish youth reproductive health counselling centres

Explanation:- Establish youth reproductive health conciliating centre is useful for population policy of Ethiopia

Answer: D

3. Which one of the following National Regional State Ethiopia has the lowest percentage urban from the regional total population?

A. SNNP
B. Somali
C. Tigray
D. Afar

Explanation:- according to the 2007 census, the SNNP has only 10% unban from the regional total popn

Answer: A

4. Which one of the following describes the socioeconomic characteristic of Ethiopia?

A. The majority of Ethiopia population has access to clean water.
B. Currently, Ethiopia is one of the middle income countries.
C. The illiteracy rate has been decreasing significantly since the 1990s.
D. More than a half of the rural population has access to electricity.

Explanation:- Since 1996, the illiteracy rate has decreased by 50%

Answer: C

5. One of the following condition can be taken as evidence of a promising future for socioeconomic development in Ethiopia. Which one is it?

A. Availability of various natural resources

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B. Diverse nature of imported goods
C. High rate of rural-to-urban migration
D. Rapid rate of population growth

Answer:

6. Why has the effectiveness of Ethiopia's plan for Accelerated and sustained Development to end poverty (**PASDEP**) remained below the anticipated level? This is due to
   A. Limited road network and infrastructure.
   B. The diverse afro-climatic nature.
   C. Shortage of human resource.
   D. High dependency on foreign aid.

Answer:

7. What kind of measures do you recommend for water resource management and conservation in Ethiopia?
   A. Planting trees on the upper parts of the streams
   B. Rain water harvesting wettest parts of the country
   C. Building dams to generate hydroelectric power
   D. Planting exotic plants near the streams and rivers

Answer:

8. If you travel from Addis Ababa to the north eastern lowlands of Ethiopia in summer, you can recognize a high variation in the rainfall. This variation is mainly caused by
   A. Distance from the sea.
   B. Latitude
   C. Mountain barriers
   D. Revolution of the earth

Explanation:- the most wind that comes from Atlantic ocean is blocked by the north-central massifs There from the Northeaster low lands of Ethiopia are under shadow (rain shadow)

Answer: C

9. Which meteorological station of Ethiopia is expected to receive rainfall throughout the year?
   A. Debrmarkos B. Gode C. Moyalle D. Bonga

Answer:

Explanation:- the region has more rainy days than any other part of

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country.

Answer: D

10. Which wild animal is unique to the Bale Mountains?
A. Mountain Nyala
B. Walia Ibex
C. Wild ass
D. Gelada baboon

Explanation:- Bole mountains are serves as a home for various endemic animals of Ethiopia, including mountain Nyala

Answer: A

11. Which of the following actions should be taken first to conserve natural vegetation in Ethiopia?
A. Practicing afro-forestry
B. Controlling overgrazing of range lands
C. Creating community's awareness
D. Establishing national parks

Explanation:- Providing env'tal education to enhance public awareness about the use management of natural vegn can be suggested for preserving Ethiopia's forests

Answer: A

12. Which type of soil found in the desert and semi-desert areas of Ethiopia?
A. Yermosols
B. Regosols
C. Acrisols
D. Cambisols

Explanation:- Yermosols are found in desert & semi desert part of Ethiopia

Answer: A

13. What is the soil conservation measure that involves resting of arable for a limited period of time called?
A. Check dams
B. Inter cropping
C. Fallowing
D. Limiting grazing

Explanation:- fallowing is soil conservation method used for limited period of time

Answer: C

14. Which of the population theories below considers the loss of human life through drought as a positive check to control fast population growth?
A. Marxian
B. Anti - Malthusian

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C. Boserupian D. Neo-Malthusian

Explanation:- Elements reffered to the catastrophes as "positive Checks" that control fast population growth

Answer: D

15. Which one of the following explains the trend of population growth rate of Ethiopia?

A. At present, Ethiopia is the third most populous country in Africa.

B. Before the 1930, the rate of population growth was very high.

C. Currently, the rate of population growth shows a slight decline.

D. Between 1989 and 2007, there was a fast population growth rate.

Explanation:- Between the three censuses, the popn of Ethiopia increased from 42.6 million in 1984 to 53.5 million in 1994 and to 73.8 million in 2007

Answer:- D

16. In which parts of Ethiopia do you mostly observe permanent rural settlements?

A. In the crop farming areas C. In the rift valley sections

B. A long the major river valleys D. In the pastoralist areas

Explanation:- The permanent rural settlements of Ethiopia are mostly associated with the crop farming highland areas of Ethiopia

Answer: A

17. If you travel from Ethiopia to Botswana in January, what type of climatic condition would you experience?

A. Spring rainfall C. Autumn rainfall

B. Summer rainfall D. Dry winter season

Explanation:- Summer rainfall because the months of December, January and February constitutes he summery seasons in the southern hemisphere.

Answer: B

18. Under what climatic regions of Africa are areas that have high temperature heavy rainfall throughout the year categorized?

A. The Tropical monsoon climate

B. The Tropical savanna climate

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C. The Mediterranean climate
D. The Tropical continental climate
Explanation:- Equation climate

Answer: B

19. What makes the northern part of Senegal and the southern part Mauritania similar?
A. Earthquake
B. Recurrent drought
C. Volcanism
D. Air pollution
Explanation:- These parts of the countries border sahel region (one of the extremely drought - affected areas in Africa)

Answer: B

20. Which of the following river is known for having the largest waterfall in Africa?
A. The Zambezi River
B. The Congo Rive
C. The Nile Rive
D. The Volta River
Explanation:- Because the Zambezi rive basin has Africa's largest waterfall Victoria fall

Answer: A

21. One of the following lakes belongs to Uganda and the Democratic Republic of Congo. Which one is it?
A. Lake Tanganyika
B. Lake Turkana
C. Lake Albert
D. Lake Chad
Explanation:- Lake Alert is belongs to Uganda and DR Congo

Answer: A

22. One of describes the economic uses of the Africa rivers?
A. They are highly navigable throughout the year
B. They are sources of construction materials.
C. They are less suitable for hydroelectric power generation
D. They have been highly utilized for irrigation purposes.
Explanation:- Rivers of African containg construction materials such as sand and gravel.

Answer: B

23. What is the foundation of the hydro politics of the Nile river?

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A. The degree to which the Nile river is utilized for different purposes in the upper and lower stream countries.

B. The drought frequency, duration and intensity expected t occur in the Nile basin countries.

C. The amount of budget that the basin countries are contribution to manage the physical environment of the Nile basin.

D. The amount of water that the upper and lower stream countries are contributing to the Nile River.

Explanation:- The degree to which Nile river is utilize for diffrent purpose in upper & lower stream counters is foundation of hydro poetics of vile river

Answer: A

24. Which one of the following wild animal of Africa is highly hunted for economic reasons?

A. Gorilla B. Elephant C. Zebra D. Oryx

Explanation:- In many parts of Africa, people hunt elephants for their tusks.

Answer: B

25. Why are some rare animals of Ethiopia approaching extinction? This is due to the

A. Extreme fight among the animals

B. Widespread practice of legal hunting

C. Prevalence f harsh climatic conditions

D. Rapid expansion of settlements.

Explanation:-Because in Ethiopia there is high popn growth rate and in turn it is reason for the expantion of settlements by deforesting reign areas. so that the wild animals can migrate and extinct.

Answer: D

26. Which of the following soils of Africa correctly paired with respect to their characteristics?

A. Calcisols ----- rich in humus

B. Acrisols ----- acidic and shallow

C. Vertisols ----- shallow and stony

D. Fluvisols ----- black basaltic

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Explanation:- Acrostic are acidic and they have shallow profile

Answer: B

27. Among the following major types of rock, which one is formed as a result of compation and cementation?

A. Metamorphic
B. Sedimentary
C. Igneous
D. Volcanic

Explanation:- Sedimenfary rocks, sedm-rocks are fromed when a small particles of matter or sedments of weathered rocks are deposited and comented.

Answer: D

28. What kind of soil is characterized by very small particles and low porosity?

A. Silt soil B. Loam soil C. Sandy soil D. Clay soil

Explanation:- Clay soils, b/c they are formed from very small particles, which are easily parcked togerther.

Answer: D

29. Why is the koppen system of climate classification more widely used than the Ancient Greeks' classification?

A. The koppen system considers the distribution of water bodies.
B. It is based on temperature and sunshine distribution.
C. It considers the averages of temperature and precipitation.
D. Its climatic region is within latitudinal boundaries.

Explanation:- Koppen climate classification uses average rainfall and tempratures

Answer: C

30. How dose continental drift cause climate change?

A. By changing the aspect and position of the landmass
B. By emitting large volume of gases and dust particles
C. By transferring high amount of heat across the globe
D. By altering composition of carbon dioxide

Explanation:- Continetal drift Causes climate change by change position of land mass

Answer: A

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31. Which climatic zone of Ethiopia is the transition between the hot arid and the humid climates?

A. Bereha B. Kolla C. Dega D. Woina-Dega

Explanation:- Kolla is transitional climate b/n hot-arid & humid climate

Answer: B

32. What is the main factor that substantially influences the soil of tropical grasslands?

A. Soil leaching C. Prolonged drought

B. Soil acidity D. Water logging

Explanation:- Soil acidity is most factor that influence soil of tropical grass land

Answer: B

33. In which one of the following continents is the population doubling time very short?

A. Latin America C. Asia

B. North America D. Europe

Explanation:- The population doubting time of Latin America is only about 41 years.

Answer: A

34. What is the major factor that forces people to migrate from their places of origin in search of employment?

A. Political factor C. Social factory

B. Economic factor D. Ecological factor

Answer:__

35. Which one of the following is the main reason for the rate of urbanization in Ethiopia?

A. Urban development policy of the country

B. High rate of urban-to-urban migration

C. The presence of a large number of immigrants

D. The presence of a high rate of natural increase.

Answer:__

36. What could be generalized about the population structure of Ethiopia?

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A. In most rural areas, the number of males is greater that of females.
B. Ethiopia has the smallest size of economically active population.
C. In urban areas, the size of male population is greater than that of the female population.
D. One - fourth of the total population of Ethiopia is the elderly population.

Answer:___

37. In Action Research, which if the following steps will help you to state the assumption you want to be verified in the research?
A. Formulation hypothesis
B. Reviewing literature
C. Stating objectives of the study
D. Analyzing and interpreting data

Explanation:- Formulating hypothesis b/c hypothesis is a priliminary assumption or tentative explanation that accounts for a set of fact.

Answer: A

38. Which of the following bounds the southern part of the Horn of Africa?
A. Sudan
B. Kenya
C. The Red Sea
D. The Indian Ocean

Explanation:- Kenya because Kenya is found to the south of horn of Africa

Answer: B

39. In which one of the following countries do you expect very diverse natural resources?
A. Somalia B. Djibouti C. Eritrea D. Ethiopia

Explanation:- Ethiopia have diverse natural resource.

Answer: D

40. One of the following demonstrates the strategic relative location of Ethiopia. Which one is it?
A. It is located in northeastern Africa.
B. It is one of the middle East countries.
C. It is situated to the southeast of Asia
D. It is found in the lower Nile basin.

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Explanation:- Strategic 10 cation of Ethiopia is North east Africa

Answer: A

41. What is the advantage of compact shape of Ethiopia?
    A. High communication gap
    B. High economic integration
    C. Possessing similar type of resources
    D. Easily direct access to the sea

    Explanation:- High economic integration is advantage based on compact shape of Ethiopia

Answer: B

42. What kind of geological event occurred during the precambrian Era in the Horn of Africa?
    A. Denudation and peneplanation
    B. Transgression regression of sea
    C. Formation of the Rift Valley
    D. Intensive volcanic activities

    Explanation:- Geological event happen in horn Asrica in precamberian era is intensive volcanic activity

Answer: D

43. While Mount tsibet is located in Tigray plateau, which mountain is located in the south western highlands?
    A. Mount Analu
    B. Mount Gughe
    C. Mount kaka
    D. Mount Batu

Answer:___

44. Which land form of the horn serves as a water watershed between the Awash and the abbay river basins?
    A. The Plateau of Eritrea
    B. The North-Central massifs
    C. The plateau of Shewa
    D. The Plateau of Tigray

    Explanation:- Plateau of shows is a land form watershed between Awash & Abbay river bas in

Answer: C

45. Which one of the following bast describes the major rivers of Ethiopia?
    A. They have steep long profile.

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B. They have steady water volume.
C. Most have no outlet to the sea.
D. They have high navigation potential.

Explanation:- steep long profile is major characteristics of Ethiopian rivers

Answer: A

46. Which of the following Ethiopia lakes is classified as the Rift Valley Lake?

A. Lake Abaya
B. Lake Tana
C. Lake Hashenge
D. Lake Haik

Explanation:- Lake Abbaya is rift valley lake

Answer: A

47. In Ethiopia, when is a given settlement considered an urban settlement?

A. When the residents on different economic activities
B. When houses are made up of simple construction materials
C. When the inhabitants are primarily engaged in agriculture
D. When population size exceeds one thousand people

Explanation:- Urban settlement of Ethiopia is depend on d/t economic activities like trade, manufacturing etc.

Answer: A

48. What kind of economic activity involves the production of food stuffs and raw materials?

A. Tertiary
B. Secondary
C. Primary
D. Quaternary

Explanation:- Primary economic activities produce food staffs and raw materials

Answer: C

49. Why is most of the farming activity practiced in the highland parts of Ethiopia?

A. The available of large-scale irrigation schemes
B. The presence of vast areas of agricultural fields
C. Favourable topographic nature of the area
D. The availability of reliable rainfall

Explanation:- Most high lands of Ethiopia farming activity is based

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available rainfall

Answer: D

50. Among the following, which one is an example of renewable natural resource?

A. Minerals B. Soil C. Coal D. Natural gas

Explanation:- Renewable resource can replace themeselves in short period after they use **Example**: Soil, air, plans solar energy etc

Answer: B

51. Currently, which one of the following wood is an important source of raw material for the furniture industries of Ethiopia?

A. Bisana B. Girar C. Wanza D. Shola

Explanation:- Wanza is important raw material in our country for producing different furnitures.

Answer: C

52. Among the following minerals, which one is used as an agricultural input industries of Ethiopia?

A. Tin B. Nitrates C. Copper D. Lead

Explanation:- Nitrates is mineral used for making fertilizer

Answer: B

53. Which one of the following is considered as an economic impact of HIV/AIDS in Ethiopia?

A. Family breakup
B. Feeling of guilty conscience
C. School dropout
D. Loss of skilled labor

Explanation:- Economic impact of HIV/AIDS is Ethiopia is loss of skill man power.

Answer: D

54. If you were a member of your school environmental conservation club, which of the following would you advice the community to do?

A. Protecting water bodies from disposal of waste materials
B. Encouraging extraction of forest resources for furniture purposes
C. Promoting free grazing of domestic animals in national parks
D. Encouraging expansion of agricultural lands

Answer:_

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55. Which of the following explains the structure of the earth?
A. Mantle is the thinnest layer of the earth.
B. Inner core is the origin of molten magma
C. Continental crust is composed of thin granite.
D. Crust has more intense pressure than other layers.

Answer:__

56. Which physical environment of the earth is associated with the ground water potential of an area?
A. The biosphere
B. The hydrosphere
C. The atmosphere
D. The lithosphere

Explanation:- The earth's physical enviroment associated with water bodies is called **Hydro sphere**

Answer: B

57. How dose the world bank contribute to the economic development of the developing countries?
A. By enforcing trade regulations
B. By financing mega projects
C. By stabilizing currency values
D. By promoting trade barriers

Explanation:- The major contribution of world bank for economic development is financing mega projects

Answer: B

58. One of following can considered as an advantage of globalization for the developing countries Which one is it?
A. It has encouraged local consumption of domestic products.
B. It has maintained the benefits of the developing countries.
C. It has increased remittance flows to the developing countries.
D. It has ensured equality between countries of the world.

Answer:__

59. Which one of the following best describes the subject of Geography?
A. It focuses on examining man-environment interactions from spatial perspective.
B. It is especially interested in describing the shape and size of the earth
C. It focuses more on the physical features including their names and their where abouts.

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D. It studies about the distribution of human population and their economic activities.

Answer:

60. Which one of the following can be addressed through topical approach?
    A. Surveying the geography of the Antarctica continent
    B. Describing the geography of sub-Saharan African region
    C. Analyzing the hydro politics of the Nile River
    D. Studying the different regions of the world

Answer:

61. In which of the following ways dose the quantitative revolution contributed to the development of geography?
    A. By integrating the idea that geography can be used to solve socioeconomic and political problems.
    B. By advocating the idea that physical factors should be investigated from different perspectives.
    C. Through formulating theory such as human beings can change their environment using the latest technology.
    D. By integrating statistical techniques, spatial reasoning and digital system in geographical research.

Answer:

62. Which one of the following is considered as the turning point in the development of geographic information systems (GIS)?
    A. The advancement of computer technology
    B. The integration of GIS in many disciplines
    C. The advancement of quantitative geography
    D. The increasing demand of GIS from different users

    Explanation:- The development of GIS is based on advancement in quantitative approach in geography.

Answer: C

63. Which one of the following water bodies shares the longest boundary with Africa?
    A. The Mediterranean sea
    B. The Red sea
    C. The Atlantic Ocean
    D. The Indian Ocean

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Explanation:- Long boundary share of water bodies for continent Africa is Atlantic ocean

Answer: C

64. What is the distinctive feature of the central Africa?

A. The smallest in Africa
C. Highland topography
B. Direct access to the sea
D. Diverse climate

Explanation:- Central Africa is the only region by district feature of diverse climate

Answer: D

65. During which geological era did the pangea break into Laurasia and the Gondwana land?

A. The precambrian Era
C. The Paleozoic Era
B. The Mesozoic Era
D. The Cenozoic Era

Explanation:- Pangea is breakdown into laurasia & Gondwana land during mesozoic era

Answer: B

66. What makes Africa the hottest continent in the world?

A. Distance from the sea
C. Latitude
B. Ocean current
D. Altitude

Explanation:- Due to altitudunal location Africa has high temperature in the world

Answer: C

67. What is the major factor that makes the Nile valley of Egypt one of the most densely populated areas of the world?

A. Dense vegetation cover
C. Favourable climate
B. Good road network
D. Fertile alluvial soil

Explanation:- Because of fertile alluvial soil Nile valley of Egypt is highly populated

Answer: D

68. If a mineralogist from Algeria travels to Ethiopia to explore potash deposit in the Afar Region, what is this type of movement called?

A. Intra-continental migration
C. In-migration
B. Out-migration
D. Inter-continental migration

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Explanation:- Intra continental migration is takes place from one country to another country with in one continent

Answer: A

69. What is the implication of the low level of urbanization in Africa?

A. Low cost of urban transport facilities

B. Low rate of urban to urban migration

C. Low rate of rural to urban migration

D. Low living standard of urban population

Explanation:- Low level of urbanization in Africa implies low living standard of urban popn

Answer: D

70. Which one of the following describes the concept of economic growth?

A. It involves the qualitative improvement of life a country

B. It measures that well - being in human society in country

C. It describes the structural growth and technological advancement.

D. It states how fast the production of agricultural crops increases.

Explanation:-Structural growth & technological advancement indicate economic growth

Answer: C

71. What is the main reason that makes intra-continental trade to be low in Africa?

A. The diverse nature of the economy

B. Lack of political willingness

C. Absence of common currency

D. Limited access to sea transport

Explanation:- Intra - continental trade in Africa is low because of absence common currency.

Answer: C

72. In which of the following countries is the contribution of Gold to its economic development the highest in Africa?

A. Republic of South Africa

B. Democratic Republic of Congo

C. Nigeria

D. Botswana

Explanation:- Republic of south Africa contribute high gold produc-

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Answer: C

73. What kind of mining technique is manly used to extract coal and metal near the surface of the ground?

A. Open pit mining
B. Quarrying
C. Placer mining
D. Strip mining

Explanation:- Technique used to extract coal & metal near to surface is placer mining

Answer: C

74. Why is a peaceful conflict management in mineral resource exploitation areas of Africa recommended?

A. To satisfy the financial needs of the ruling party
B. To foster socioeconomic development
C. To increase mineral resource exploitation
D. To limit mineral export to the western world

Explanation:- Peaceful conflict management in mineral resource exploitation in Africa is recommanded for increasing socio economic development

Answer: B

75. In which one of the following contexts can the quantitative research approach be used?

A. Identifying the basic characteristics of a given region
B. Examining the property of soils of the Ethiopian highlands
C. Describing the differences between the Abay and Awash rivers
D. Investigating the relationship between rainfall and elevation of an area

Explanation:- Investigating numerical research approach is quantitative research

Answer: D

76. Why is geographic research different from the other research types?

A. The cause-effect relationships of different phenomena.
B. The social challenges from a temporal point of view.
C. The factors that effect the spatial distribution of population
D. The geological formation through mathematical modeling.

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Explanation:- Geographical research is focused on cause & effect relationship of diffrent phenomena.

Answer: A

77. Which branch of geography is mainly concerned with social beliefs and customs?

A. Population geography
B. Urban geography
C. Political geography
D. Cultural geography

Explanation:- Branch of geography that deals with social beliefs & customs is cultural geography.

Answer: D

78. Under what condition are fold mountains formed?

A. When the earth crust is folded by tensional force
B. When the solid materials of the earth gets bent upward
C. When the forces pull away from each other on the surface
D. When the crust makes cracks due to compressional forces

Explanation:- Fold mountain is formed by bending upward of earth crust

Answer: B

79. What type of geographic feature is formed when magma cools and solidifies before reaching the surface of the earth?

A. Batholith B. Crater C. Caldera D. Volcano

Explanation:- Geographic feature formed by magma cools & solidify before reaching the surface is called Batholith.

Answer: A

80. Which one of the following is a common agent for both physical and chemical weathering?

A. Temperature B. Frost C. Rainfall D. Plants

Explanation:- The common agents for physical & chemical weathering is plants action

Answer: D

81. Among the following, which one is a cause for the formation of barchans?

A. Running water
B. Sea Wave
C. Wind
D. Moving ice

Explanation:- Barchan sand dure & loes deposit is formed by wind erosion

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Answer: C

82. Which one of the following, explains layers of the earth's atmosphere ?
A. Stratosphere is the coldest part of the atmosphere.
B. Thermosphere is known for having extremely low density
C. Mesosphere is considered as the home of ozone gas.
D. Troposphere is a layer where meteorites burn and disintegrate.

Answer:

83. What type of wind might a visitor to one of the coastal areas in Africa observe?
A. Monsoon winds
B. Mountain breezes
C. Sea breezes
D. Planetary winds

Explanation:- winds happen in coastal areas of Africa is sea breezes

Answer: C

84. What characterizes the coniferous forest region of earth?
A. Agriculture is the most important economic activity of the region
B. The region is dominated by a single type of plant species.
C. Winter is a period of long days of bright sunshine the region.
D. The region receives a high amount of rainfall throughout the year.

Explanation:- Coniferous forest region is characterized by single type of plant species

Answer: B

85. In what way is Tundra differentiated from the polar Ice Cap region?
A. It is located both in the north and the south poles.
B. Its precipitation mainly falls in the form of snow.
C. Is is permanently covered by ice and snow.
D. Its one-third of months have a positive temperature.

Explanation:- Tundra is differentiate from polar ice caps by rain fall is happen in the form of snow

Answer: B

86. How has advancement in transport and communication technology brought changes in the location of economic activities?
A. By increasing the transportation cost of manufactured goods from

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producers to consumers.

B. By supporting industrial workers to live close to their work place.

C. By enabling producers to sell perishable goods fresh anywhere.

D. By facilitating the location of industries to be close to the source of raw materials.

Explanation:- Advancement in transport & communication technology brings producers to sell perishable products as a fresh maner in any where.

Answer: C

Station	Item	Months											
		J	F	M	A	M	J	J	A	S	O	N	D
G	Temp(°C)	15	16	17	17	17	16	15	25	15	14	15	14
	RF(mm)	18	28	72	147	178	271	386	419	286	51	11	7
H	Temp(°C)	27	27	26	26	26	25	25	25	24	25	25	26
	RF(mm)	65	96	157	157	135	103	112	163	200	205	161	89
K	Temp(°C)	9	9	12	14	17	21	25	25	22	19	13	11
	RF(mm)	103	84	86	84	64	21	7	17	52	65	88	100
L	Temp(°C)	26	27	29	32	34	36	35	33	34	31	28	26
	RF(mm)	5	7	19	25	20	3	31	63	29	13	5	4

87. Under what type of biome can station 'L' be categorized?

A. Grassland C. Deciduous forest

B. Desert D. Taiga

Explanation:- Station 'L' categorized under desert climate because temperature is high & rain is low.

Answer: B

88. Which meteorological station has the lowest mean annual temperature?

A. L B. G C. H D. K

Explanation:- The lowest mean annual temperature is observed in station "G" because when we add 12 month Temperature & divide the sum to 12 is low

Answer: B

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$$\text{Annual mean } T^0 (G) = \frac{15+16+17+17+17+16+15+15+15+14+15+14+15+14}{12}$$

$$= 15.5$$

89. What type of rainfall is most likely common in station H?

A. Convectional
C. Frontal
B. Orographic
D. Cyclonic

Explanation:- Type of rainfall observed in station "H" is conventional

Answer: A

90. Which of the following stations dominantly experiences the Mediterranean type of climate?

A. K B. H C. G D. L

Total Population	11,818,619
Male Population	5,862,035
Female Population	5,956,584
Rural Population	5,597,189
Age Dependency Ratio	43%
Live Births	196,189
Death Rate	5.5/1000
Total Area	163,610km^2
Cultivated Land	97,430km^2

Explanation:- Mediterranean climate type is observed in station "L" because of temperature & rain fall variability.

Answer: D

91. What does the age dependency ratio of 43% imply?

A. It shows 57% of the total population is dependent on 43% of the labor force.
B. It refers to 100 persons who are dependent on every on every 43 economically active persons.
C. It means 43% of total population depends on economically active labor force.

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D. It implies that 43% of the total population is economically active.

Explanation:- Based on the information ADR = 43% means 43% of total popn depends on economically active labor force.

Answer: C

92. What is the rate of natural increase of the population?

A. 1.1% B. 1.7% C. 11.1% D. 1.5%

Explanation:- $RNI = BR - DR = \frac{16.615}{1000} - \frac{5.5}{1000} 11.1$

Answer: C

93. The agricultural density of population is

A. 121.3P/km² B. 21.9P/km² C. 72.2P/km² D. 36.9P/km²

Explanation:-

$$\text{Agricultural desity} = \frac{\text{Rural popn}}{\text{Cultivated land}} = \frac{3,597,189}{97,430} = 36.9 \, P/km^2$$

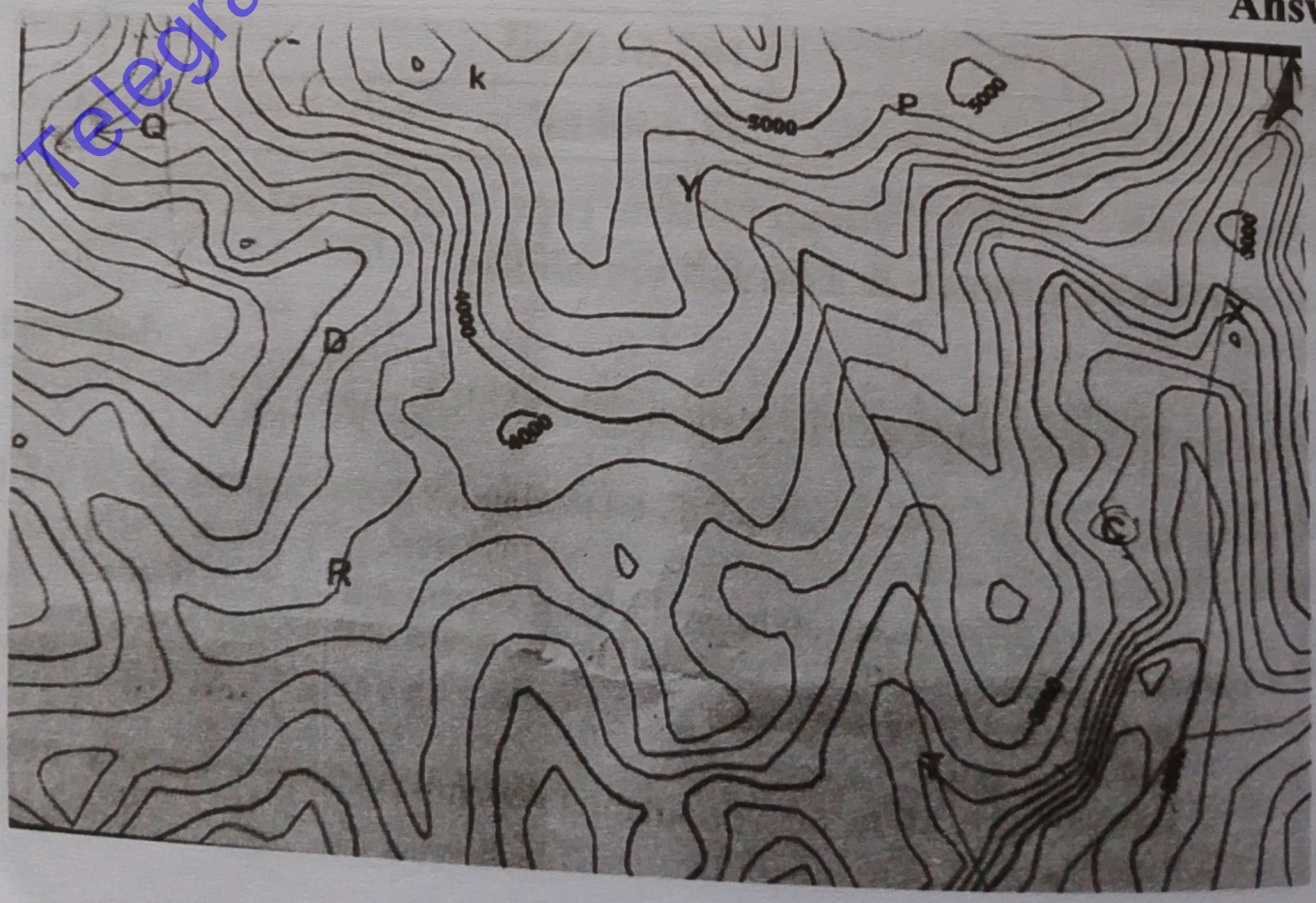
Answer: D

94. What is crude Birth Rate of the region?

A. 17.5% B. 6.6% C. 1.7% D. 16.6%

Explanation:- $CBR = \frac{B}{P} \times K = \frac{196,189}{11,818,619} \times 1000 = 16.6\%$

Answer: D



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95. What is the direction of "D" from "Q" ?
A. Southwest B. Northwest C. Northeast D. Southeast
Explanation:- Based on map direction of "D" from "Q" is south east.
Answer: D

96. What is the gradient of the slope between 'C' and 'P' (Map distance between C and P = 8cm)
A. 27.8 % B. 30.7 % C. 20.9 % D. 25.6 %
Answer:___

97. What is the areal scale of the given map?
A. 1 cm^2 to 8 km^2
B. 1 cm^2 to 2 km^2
C. 1 cm^2 to 4 km^2
D. 1 cm^2 to 1 km^2
Explanation:- Areal scale of map is 1 cm^2 to 1 km^2
Answer: D

98. What king of statistical diagram is more suitable to show the relationship between temperature and elevation in Ethiopia?
A. Histogram B. Bar graph C. Pie chart D. Line graph
Explanation:- Statistical diagrams used to show r/ship of altitude and temperature is bar graph
Answer: A

99. What type landforms is shown between 'Y' and 'T'?
A. Saddle B. Hill C. Valley D. Depression
Explanation:- land form shown in map by 'Y' & 'T' is saddle
Answer: A

100. If we reduce this map two times, what will be the scale of the new map?
A. 1: 75,000 B. 1: 100,000 C. 1: 200,000 D. 1: 50,000
Explanation:- When we reduce map scale 1: 100,000 two times w/c gets 1: 200,000
Answer: C

101. Of the following regions of Africa, which one has the lowest infant mortality rate?
A. Eastern Africa
B. Northern Africa
C. Southern Africa
D. Western Africa
Explanation:- Lowest infant mortality is recorded in Africa's region

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in North Africa

Answer: B

102. What is the impact of rapid rate of urbanization in Africa?

A. Enhancing Planned development

B. An increase in living standard

C. Environmental degradation

D. Reduction in social evils.

Explanation:- Impact of rapid urbanization in Africa is environmental degradation

Answer: C

103. How dose cloud cove cause variation in the spatial distribution of air temperature across the horn of Africa?

A. By deflecting the direction of the incoming solar radiation

B. By regulating the incoming and outgoing solar energy

C. By modifying the angle of the overhead sun

D. By allowing the full force of the solar energy to reach the earth

Explanation:- Cloud cover affects spatial distribution of air temperature in horn Africa by regulate incoming solar energy

Answer: B

104. What makes the lowland part of Ethiopia sparsely populated?

A. Low temperature

B. Poor soil fertility

C. Prevalence of malaria

D. Low Plant species

Explanation:- Lowland of Ethiopia is sparsely populated because of presence in epidemic disease like malaria.

Answer: C

105. Among the following areas, which one has the highest crude population density in Ethiopia?

A. Gedeo B. Wolayita C. Assosa D. Warder

Explanation:- Highest crude popn density in Ethiopia is mostly in conducive climatic areas like Gedeo

Answer: A

106. Which continent shares the smallest percentage the total area of the world?

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A. Europe
B. Australia
C. North America
D. South America

Explanation:- Continent shares smallest percentage of total area of world is Australia

Answer: B

107. Africa is considered as a developing continent because of the following reasons. Which one is it?
A. Africa's economy is based on export-oriented mineral resources.
B. The economy of Africa is highly dependent on primary activities.
C. High proportion of the population are engaged in service sector.
D. There is relatively uniform distribution of wealth in the continent.

Answer:___

108. If you conduct geographic research, in which part of the research would you discuss the statistical techniques to be employed in the study?
A. Data Analysis
B. Research design
C. Research Hypothesis
D. Data collection and organization

Explanation:- In geographic research to conduct quantitative techniques is used to analyses data

Answer: A

109. Which statement describes the relief type of rainfall?
A. It is common in humid parts of the world where the temperature is high throughout the year.
B. It is the most common type of rainfall in the lowland parts of world
C. It brings high amount of rainfall to the leeward side of the mountains.
D. It is formed when warm and moisture laden wind is forced to ascend to the top mountains.

Answer:___

110. Which one of the following describes the equatorial rainforest region?
A. It is mostly inhabited by arboreal animals
B. It forms a continuous belt around the earth
C. It is dominated by softwood trees
D. It has a high annual range of temperature.

Answer: ___

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111. Among the following, which one explains the economy of Ethiopia?
A. The construction sector is the leading sector in terms of GDP share.
B. The service sector is currently the dominant sector.
C. The industrial sector employs more than a half the labor force.
D. There is a strong link between various economic sectors.

Answer: ___

112. What makes Ethiopia advantageous to strengthen its economic relationship with the rest of the world?
A. Geographical proximity to other continents
B. The high quality of its industrial products
C. Large areal size of the country
D. The large size human population

Explanation:- Ethiopia is advent a geous to strength economic r/ship with other world is due to geographical proximity to other countries

Answer: A

113. Which one of the following is a direct cause for the occurrence of drought in Africa?
A. Topographic nature
B. Distance from the sea
C. Volcanic eruption
D. Climate variability

Explanation:- The direct cause of drought in Africa is climate change

Answer: D

114. What kind of soil is found in the Mediterranean climatic of Africa?
A. Luvisols B. Calsisols C. Acrisols D. Regosols

Explanation:- Type of soil found in merited anean climate region of Africa is Luvisols.

Answer: A

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SECTION 6 – SCHOLASTIC APTITUDE

HOW TO WORK IT OUT

1. **Aptitude Test**
   - **What is aptitude?** It is natural ability or skill at doing something. Its synonym is talent. **(Aptitude = talent).**
   - **What is aptitude Test?** It is one designed to show whether somebody has the natural skill or ability for a particular job or course of education.
   - **Aptitude Test** such as **Scholastic Aptitude Test** is a college university entrance examination that measure basic skills used in many different subject areas or **achievement tests** that measure knowledge of specific subject such as history or biology.
   - **Scholastic Aptitude Test** measure skills that are gained over a period of many years. Consequently, it is impossible to study for them in the usual sense. In other words, you can't simply sit down a few days before these exams and learn everything that you need to know for them. Instead, you must begin now to develop the skills that these tests measure, including **vocabulary,** your **reading** and **computational skills;** your **ability** to think **reasonably** and **critically**; and your **understanding** of the basic principles of English.
   - The **Scholastic Aptitude Test (SAT)** is divided into sections on **Verbal** and **Mathematical** ability.
   - The types of English items included in **scholastic aptitude test** are called **VERBAL REASONIGN**.
   - To prepare for this college entrance examination (SAT), familiarize yourself with the types of questions that the **verbal reasoning** contains.

   ☞ These include:

      1. Analogy
      2. Synonyms
      3. Antonyms
      4. Language usage
      5. Grammatical meaning
      6. Sentence completion
      7. Reading comprehension
      8. Logical reasoning

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9. Analytical reasoning

2. Analogy

- **Analogy** are words that are related to each other in some way. In an analogy question, you are given two words that are related, and you must determine what this relationship is. You then choose a second pair of words that are related in the same way.
- A typical analogy question looks like this.

Example: Analogies

SONG: REPERTOIRE

A. Score: Melody
B. Solo: Chorus
C. Instrument: artist
D. Suit: wardrobe
E. Benediction: church

- These are strategies to use in answering analogy questions.

1. First determine the relationship between containing **both of them.**
2. Then, find a second pair of words that you can substitute for the original pair.
   Several **suits** make up a **wardrobe.**

- Common relationships found in analogy questions include.

Types of analogy	Examples
Action to object	Play: clarinet
Cause to effect	Sun: sunburn
Item to category	Iguana: reptile
Object to its function	Pencil: writing
Object to its material	Curtains: cloth
Part to whole	Page: book
Time sequence	Recent: current
Word to antonym	Help: hinder
Word to synonym	Provisions: supplies
Worker and creation	Artist: sketch
Worker and tool	Lumberjack: saw
Type of characteristic	Dancer: agile

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1.2 Synonyms (SYN)

- **Synonyms** are words that have similar to meanings. Most **synonyms** don't mean exactly the same thing knowing the small differences in meaning between two synonyms enables you to each word precisely.
- Learning **synonyms** will help you broaden your vocabulary, expanding the range of words available to you and enabling you to express your ideas with precision.

Examples: Sincere, unaffected, heartfelt

- ☞ Show, display, exhibit, expose
- ☞ Expect, anticipate, hope, await
- ☞ Contain, hold, accommodate
- ☞ Rational, reasonable, sensible
- ☞ are synonym words

1.3 Antonyms

- An **antonym** is a word that is opposite in meaning to another word. **Antonym questions** provide a single word and ask you to select from a list of words the one that is most opposite meaning.
- A typical antonym question looks like this:

Example: Antonyms
DEPRESS:

A.	Force	C.	Clarify	E.	Loosen
B.	Allow	D.	Elate		

- To answer an antonym question, use the following strategies.

1. Look for a word that is opposite in meaning. Don't be thrown off by any synonyms – words that are similar in meaning – that included among the choices.
2. Before looking at the choices be sure that your know the meaning of the first word. Define it using any of the following methods:
   a. Think of another word or group of words that means the same thing.
   b. Think of a sentence or sentences that use the word, then try to arrive at an exact definition.
   c. Try analyzing the parts of the word.

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3. Once you know the meaning of the first word, look at the choices. If none of these is obviously the correct antonym, try either or both of the following strategies.
   a. Eliminate any odiously correct answers.
   b. Remember that many words have more than one meaning. If none of the choices seems to be opposite in meaning, think of other meanings of the first word.

- **Therefore**, the answer to the sample question is **(D)**.

14 Language usage questions (usage questions)

- This kind of question can ask either the corresponding meaning or phrases or idioms or the grammatical error within the same sentence.
- **Idioms (especial meaning phrases)** can be done through the contextual meaning they have in a sentence or we need to know the especial expressions they originally have.
- **Grammatical usage (seeking error)** question require that you recognize wiring that is not standard written English. In usage questions, four words or phrases are underlined and lettered. You must either choose the underlined part the contains an error or mark the answer corresponding to no error.
- Here is a typical usage question.

> **Example: Usage questions**
> He spoke blunty and angrily to we spectators. No

- Use the following strategies to answer usage questions.

1. Read the sentences in its entirety.
2. Look for errors of the following kinds.
   a. Incorrect word error
   b. Errors in punctuation
   c. Errors in capitalization
   d. Errors in grammar
   e. Improper, awkward, or unclear sentence structure
3. Common grammatical errors found in usage questions include the following: incorrect verb tenses, improper agreement of pronoun and antecedent or of subject and verb, improper form of the pronoun, and use of an adjective for an adverb or an adverb for an adjective. Common errors in sentence structure include **fragments, dangling modifier,**

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improper parallelism, and run – on caused by comma faults.

4. Remember that the error, if there is one, appears in one of the underlined sections of the sentence.

- **Thus,** the correct answer to the example question is (c). The object pronoun '**us**' should follow the preposition.

1.5 Grammatical Meaning questions

This is the same as/ to mean language usage. This type of question is to ask you to choose a sentence that carries the closest/ nearest meaning to the given sentence (with original sentence).

1.6 Sentence completion questions

- **Sentence completion** questions measure both your vocabulary and your ability to recognize relationships between parts of a sentence. You are given a sentence from which one or two words have been removed. You must choose words that best fit the meaning of the rest of the sentence.
- A typical sentence completion question looks like this.

Example: Sentence completion

The author was determined to _____ his own conclusion, so he _____ any information that didn't support it.

A. Unhold . . . ignored
B. Revise . . . destroyed
C. Advance . . . devised
D. disprove . . . destroyed
E. reverse . . . confiscated

- To answer sentence completion questions, use the following strategies.

1. Read the incomplete sentence carefully, noting any key words. Pay particular attention to words that indicate contrast (**but however, on the to her hand**) or similarly (**and, another, the same as**).
2. Try each of the choices in the sentence, eliminating any that make no sense or that are grammatically incorrect.
3. Don't misled by answers that contain only one word that fits well into the sentence. Incorrect answers often contain one word that makes sense and another one that doesn't. In the example, the first word in answer c makes sense, but the second word doesn't. Answer **A** is correct.

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4. Look for grammatical clues in the sentence. Does the sentence call for a verb, an adverb, a noun? If a verb is required, what should it tense be? By asking such questions, you can sometimes eliminate incorrect answers.

1.7 Reading comprehension question

- The **reading comprehension** questions measure your ability to understand various kinds of reading passages. The passages are taken from such fields as **biological** and **physical sciences, the humanities** and **the social science.**
- To answer **reading comprehension** questions, use the following strategies.

1. Expect questions on the following:
   a. The central thought of the passage.
   b. Specific details in the passage.
   c. Conclusions that can be made based upon information provided in the passage.
   d. The meaning of words in the passage.
   e. The mood of the passage.
   f. Specific techniques used by the writer of the passage.
   g. Relationships between things in the passage.
2. As you read the passage, identify the main idea and pay particular attention to transitional and other words that show relationships.
3. Read all the choices before you select an answer.
4. Answer the questions based on material in the passage, not on your own knowledge or opinions.
5. Don't choose an answer simply because it is true. Make certain that you answer the questions being asked.
6. Don't be misled by answers that are partially correct.
7. Budget your time. Set a time limit for reading each passage and answering questions on it. Don't waste time on impossibly difficult questions. Refer to your watch to make sure that you are not spending too much time.

1.8 Logical reasoning question

- Logic is the study of **reasoning (logical reasoning).**
- One of the most basic types of **reasoning** is known as **induction.**

A. Inductive reasoning

- This is the process arriving at a general conclusion based up on **specific facts**.

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A simple example of **induction** may be illustrated as follows.

1. First, you observe several specific facts. In the past, candidate smith has always voted in favor of federally – funded jobs programs. In a speech before the league of women voters, candidate smith expressed his support for federally – funded any care centers. In a new conference, candidate – smith stated that he plans to introduce legislation that will provide federal assistance to pay the utility bills of people who need help.
2. Then, having noticed a similarity between these specific facts, you draw a conclusion.
   - ☞ Candidate smith usually supports federally – funded social programs.
     The conclusion that is drawn through inductive reasoning is called **generalization**. The group of specific on which the generalization is based is called a **sample**.
   - ☞ [specific fact + specific fact → generalization]

B. Deductive reasoning

- In **deductive**, you start with a generalization and then derive a specific fact from it.

[Generalization + related fact → specific fact]

E.g.1 Snakes that have fangs are poisonous (generalization). The asp viper has fangs (**related facts**). Therefore, the asp viper is poisonous (specific fact).

Example 2: All ballads are poems (**Major premise**)
"Barbara Allen" is a ballad. (**Minor premise**)
Therefore, "Barbara Allen" is a poem. (**Conclusion**)

1.9 Analytical Reasoning questions

- **Analytical reasoning** is logic that is inferred through the virtue of the statement's own content, and it can be used to solve many types of problems in almost any discipline. This analytical aptitude test assesses inductive and deductive reasoning skills.

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SECTION ONE: Verbal Reasoning

This section contains a total 35 verbal reasoning questions with subsections of analogy, synonyms, antonyms, classification, reading comprehension, advanced verbal anagrams, analytical and logical reasoning. Each sub-section has its own directions.

Analogy

Directions: to answer questions 1-3, read the related pair of words written in capital letters followed by four pairs of words. Select the pair that best expresses a relationship similar to that expressed in the original capitalized pair. Then find the corresponding number on the separate answer sheet and blacken the letter of your choice.

1. NATIVE: FOREIGN
   A. MEAN: UNKIND
   B. DOMESTICE: HOMEGROWN
   C. LOCAL:IMPORT
   D. NAÏVE:SOPHISTICATED

   Clue: The two words have opposite relation ship
   Naive = someone who lacks knowledge and experience
   Sophisticated: some who is full of knowledge and experience
   Answer: D

2. GALLON:LITRE
   A. KILOMETRE:METRE
   B. YARD: POUND
   C. MILE:KILOMETRE
   D. POUND: WEIGHT

   Clue: Gallon is a unit of volume approximately 3-785 Littre
   Pound is of mas approximately 373-242g **Answer: D**

3. BELT: WAIST
   A. WATCH: ARM
   B. BRACELET: WRIST
   C. SHOE: LEG
   D. STOCKING: GLOVE

   Clue: Belt is a band worn around the waist
   Bracelet is a band/chain worn around the wrist. **Answer: B**

Direction: For questions 4-7 read each of the following words carefully and choose the word that correctly completes the analogy. Then blacken the letter of your choice on the separate answer sheet provided.

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4. MOUNT is to HORSE as _____ is to TRAIN.
   A. ride B. drive C. lift D. board
   Answer: B

5. HIRE is to SELECT is to _____.
   A. Reject B. Employ C. Admit D. Receive
   (opposite meaning)
   Answer: A

6. QUIET is to NOISY as CALM is to _____.
   A. Dormant B. Slow C. Agitated D. Arrogance
   (opposite meaning)
   Answer: D

7. WAR is to DESTRUCTION as GERM is to _____.
   A. Disease B. Bacteria C. Doctor D. Hospital
   (Cause and effect relationship)
   Answer: A

Synonyms

Directions: Questions **8-10** consist of words printed in capital letters followed by four words. Choose the word that is most nearly **similar** in meaning to the word in capital letters. Then, find the question number and **blacken** the letter of your choice on the separate answer sheet provided.

8. INDEPENDENCE
   A. Individualism C. Cordiality
   B. Colonialism D. sovereignty
   Answer: D

9. PERSEVERANCE
   A. Lack of energy C. Continuity of effort
   B. Frequent suffering D. Severity of disease
   Answer: C

10. EXTINGUISH
    A. Ignite B. Switch off C. Distinguish D. Turn on
    Answer: B

Antonyms

Directions: Questions **11-15** consist of words printed in capital letters followed by four words. Choose the word that is most nearly **opposite** in meaning to the word in capital letters. Then, find the

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question number and **blacken** the letter of your choice on the separate answer sheet.

11. FOLLOW
A. Go after B. Trail C. Go behind D. Precede
Answer: A

12. EXTRAORDINARY
A. Amazing B. Unique C. Unexpected D. Usual
Answer: D

13. DOMESTIC
A. Local C. Household
B. Foreign D. Neighborhood
Answer: B

14. SINCERE
A. Dishonest B. Dissent C. Genuine D. Disapprove
Answer: A

15. FRAUD
A. Scam B. Fake C. Authentic D. Fragile

Classification items

Directions: In questions **16-18**, three out of the four words are in the same classification. Choose the word that does not belong in the same classification as the others and blacken the letter of your choice on the separate answer sheet provided.

16. Choose the odd one out.
A. Disobey B. Disregard C. Respect D. Ignore
Answer: C

17. Which one of the following words is different from the others?
A. fragile B. helpless C. unbreakable D. weak
Answer: C

18. Which of the following words has a different meaning from the others.
A. Shade B. bright C. shining D. luminous
Answer: A

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Reading comprehension

Directions: 19-23 deal with restoring words correctly into a passage. The following four words listed at removed from the passage given below. Numbers are written instead of the words. Read the passage and choose the word that correctly completes the passage. Then, blacken the letter of your choice on the separate answer sheet provided.

brain potential encouragement creative specialization

We all have the potential to be1, however, because of the pressures of modern living and the need for 2, many of us never have the time or opportunity, or indeed are given the 3, to explore our latent talents, even though most of us have sufficient ammunition to realize this 4 in the form of data which has been fed into, collated and processed by the 5 over many years.

19. **Which of the following is the correct word that should replace 5(the fifth blank in the passage)?**

A. brain B. potential C. creative D. specialization

Answer: A

20. **Which of the following is the correct word that should replace4 (the fourth blank in the passage)?**

A. brain B. creative C. potential D. specialization

Answer: C

21. **Which of the following is the correct word that should replace 1(the first blank in the passage)?**

A. creative C. encouragement

B. specialization D. potential

Answer: A

22. **Which of the following is the correct word that should replace 3 (the third blank in the passage)?**

A. brain C. encouragement

B. creative D. specialization

Answer: A

23. **Which of the following is the correct word that should replace 2(the second blank in the passage)?**

A. Potential B. specialization C. creative D. brain

Answer: B

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Advanced Verbal Test Anagrams

DIRECTIONS: Questions 24-25 deal with word anagrams. Read each question carefully and give your answer by blackening the letter of your choice on the separate answer sheet provided.

24. Only one the groups of seven letters below can be rearranged to yield a seven-letter English word which is an item. Identify.

A. visible B. success C. recital D. crooked

Answer: C

25. Only one of the groups of four letters below can be rearranged to spell-out a four letter English word which is a direction

A. norm B. inch C. tank D. seat

(seat = east) **Answer: D**

Analytical and Logical Reasoning Questions

Directions: Answer questions (26-29) based on the following information. Then, blacken the letter of your choice on the separate answer sheet provided.

The director of a small rural primary school plans to assign six teachers – Kargebo, Melaku, Nahom, Paulos, Robel, and Samson – tutor/ help students after school from Monday to Saturday according to the following rule:

Melaku must be before Robel
Paulos must tutor on Tuesday
Kargebo must tutor on the day immediately before or after Nahom tutored

26. If Melaku is assigned on Wednesday, Robel cannot be assigned on:

A. Thursday B. Monday C. Friday D. Saturday

Answer: B

27. If Nahom is assigned on Saturday, When should Kargebo be assigned?

A. Saturday B. Wednesday C. Thursday D. Monday

Answer: A

28. If Nahom is assigned on Wednesday, Kargebo is to be assigned:

A. Saturday B. Thursday C. Friday D. Monday

Answer: B

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29. Which can be assigned on Monday?

A. Kargebo B. Robel C. Paulos D. Melaku

Answer: A

Directions: Read each questions (30-32) below and choose the correct answer. Then, blacken the letter of your choice on the separate answer sheet provided.

30. A factory worker has one more than five children No one else in the factory has six children. Which of the following conclusions logically follows from the given statements?

A. All workers in the factory have six children each.

B. Some of the factory have more than six children

C. Everybody in the factory has children **Answer: C**

31. According to one psychological theory, in order to be happy, one must have an intimate relationship with another person. Yet the word's greatest composers spent most of their time in solitude and had no intimate relationships. So the psychological theory must be wrong.

The conclusion above assumes that

A. The world's greatest composers chose to avoid intimate relationship

B. People who have intimate relationships spend little time in solitude

C. The world's greatest composers were happy

D. Less well known composers had intimate relationship

Answer: C

32. Adama is one the cities with lowest car accidents, and it is one of the few cities that require strict annual car safety inspections. Therefore, all cities of Ethiopia should apply similar safety inspection procedures.

The argument above is based on which of the following assumptions?

A. Annual car safety inspections contribute significantly to reduce Adama car accident.

B. Most cities are doubtful that annual car safety inspection can reduce accidents

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C. Most cities tend to impose few rules on the cars drivers use
D. Smaller number of accidents can be reported in other major cities of Ethiopia

Answer: A

Direction: Answer questions 33-35 based on the information given the statement and conclusions that follow

33. **Statement:** North Korea faced growing international opposition for its decision to explode nuclear weapons at its test site.

Conclusions: (I) the citizens of the nation favored the decision.
(II) some powerful countries do not want nations to become powerful as they are.

A. Neither conclusion (I) nor (II) follows
B. Both conclusion (I) and (II) follow
C. Only conclusion (I) fallows
D. Only conclusion (II) follows

Answer: D

34. **Statements:** Population increase coupled with recurrent drought I going to be the scenario of many Africa countries in the days to come.

Conclusions: (I) The population of developing countries will not continue to increase in the future
(II) It will be very difficult for the governments of many African countries to provide its people decent quality of life

A. Neither conclusion (I) nor (II) follows
B. Only conclusion (I) follows
C. Both conclusion (I) and (II) follow
D. Only conclusion (II) follows

Answer: D

35. **Statements**: The teacher humiliated Abraham in the presence of this colleagues.

Conclusion: (I)The teacher did not like Abraham.
(II) Abraham was not popular with his colleagues.

A. Only conclusion (I) follows
B. Both conclusions (I) and (II) follow
C. Only conclusions (I) and (II) follows
D. Neither conclusion (I) nor (II) follows

Answer: D

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SECTION One: Quantitative Reasoning (Questions 1-25)

Direction: Solve each of the problems given in questions 1-17 and give your answer by blackening the letter of your choice on the separate answer sheet provided.

1. Find the value of X in the following sequence.

6,9,7,10,8,11,9, 12, x...

A. 15 B. 14 C.10 D. 9

Solution:

Given: 6, 9, 7, 10, 8, 11, 9, 12, x, ...

Divide the given sequence into two other sequences

$\{6,7,8,9,x,....\}$ and $\{9,10,11,12,....\}$

In this case, the two new sequences are arithmetic sequence with d = 1.

Therefore, $x = 10$ **Answer:- C**

2. Find the value of X in the following sequence?

16, 8, 4, 2, x, $\frac{1}{2}$,...

A. 1 B. 2 C. $\frac{1}{4}$ D. $\frac{1}{8}$

Solution: Given: 16, 8, 4, 2, x, $\frac{1}{2}$,...

The sequence forms a geometric sequence with common ratio $r=\frac{1}{2}$.

Then, $G_5 = \frac{1}{2}G_4 = \frac{1}{2}.2 = 1$ **Answer:- A**

3. Which of the following numbers is different from the others in the group?

A. 37 B. 29 C. 27 D. 17

Solution: All the given numbers are prime numbers except 27.

Answer:- C

4. Which of the following numbers pair is different from the others in the group?

A. 7, 3 B. 6, 2 C. 3, 5 D. 1,7

Solution: All the given pair of numbers are relatively prime GCF(a,b) = 1 except the pair 6,2. **Answer:- B**

5. What is sum of the first five prime numbers? (S)

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A. 28 B. 26 C. 18 D. 11

Solution: The first five prime members are: 2, 3, 5, 7, and 11
Then, $2+3+5+7+11=28$ **Answer:- A**

6. If ABCD is a four digit positive number divisible by 5 and 10, what is the value of D?

A. It can't be determined B. 1 C. 2 D. 0

Solution:
Let 'ABCD' be a four digit number.
i) If it is divisible by 5, then $D=0$ or $D=5$.
ii) If it is divisible by 10, then $D=0$.
Since ABCD is both divisible by 5 and 10, then D = Od **Answer:- D**

7. What is the length of a side a square with area c?

A. $\sqrt{c}$ B. 2c C. 4c D. c^2

Solution: Let ABCD be a square with side x. Then, a (ABCD) = x^2
$C=x^2 \quad \Rightarrow x=\sqrt{C}$ **Answer:- A**

8. What is the value *X* in the following sequence?

10, 15, 22, *x*, 42, 55, 70…

A. 28 B. 29 C. 30 D. 31

Solution:
Given: 10, 15, 22, *x*, 42, 55, 70,… Then, the difference between consecutive terms becomes: 5, 7, a, b, 13, 15, … where $a = x - 22$ and $b = 42 - x$. This new sequence forms all arithmetic progression with $d = 2$ (i.e $d = 7 - 5 = 15 - 13 = 2$)
$\Rightarrow a - 7 = 2 \quad \Rightarrow a = 9.$
Thus, $a = x - 22$
$9 = x - 22 \quad \Rightarrow \quad x = 31$ **Answer:- D**

9. Find the value of *X* in the following sequence.

3, 12, X, 48, 75…

A. 15 B. 21 C. 27 D. 36

Solution:
Given: 3, 12, *x*, 48, 75… Then,
$3 = 3 \times 1 = 3 \times 1^2$
$12 = 3 \times 4 = 3 \times 2^2$
$x = ? \quad = ?$
$48 = 3 \times 16 = 3 \times 4^2$

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$75 = 3\times25 = 3\times 5^2$

By the above relation, $x = 3\times3^2 = 27$ **Answer:- C**

10. If one-third of one-fourth of a number is 15, then what is the three-tenth of that number?

A. 35 B. 36 C. 45 D. 54

Solution:

Let x be the required number.

$\frac{1}{3}\left(\frac{1}{4}x\right)=15$ and $\frac{1}{12}x=15 \Rightarrow x=12\times15=180.$

Then, $\frac{3}{10}x=\frac{3}{10}\times180 = 54$ **Answer : D**

11. What number should replace *X* in the following series?

-1, 0, 2, 6, *X*, 30, 62,…

A. 24 B. 14 C. 12 D. 10

Solution:

Given: -1, 0, 2, 6, *X*, 30, 62,…

The difference between consecutive terms becomes: 1, 2, 4, a, b, 32, … where $a = x-6$ and $b = 30-x$ This new sequence forms a geometric sequence with $r=2$ (i.e $\frac{2}{1}=\frac{4}{2}=2$)

$\Rightarrow \frac{a}{4}=2 \Rightarrow a = 8.$

But, $a=x-6 \Rightarrow 8=x-6 \quad x=14$ **Answer:- B**

12. How many irrational numbers are there between 2 and 7?

A. 10 B. 20 C. 30 D. infinitely many

Solution:-Fact: Between two consecutive integers, there are infinitely many rational and irrational numbers.

Answer:- D

13. The square of a positive number is greater than the number itself when the number is:

A. greater than zero
B. greater than one
C. between zero and one
D. impossible to determine

Solution:

Let X*(Equation)* be a positive no and $x^2 > x$.

Then, $x^2 > x \Rightarrow x^2 - x > 0$

$\Rightarrow x(x-1) > 0$

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$\Rightarrow x > 0$ and $x - 1 > 0$ or $x < 0$ and $x - 1 < 0$

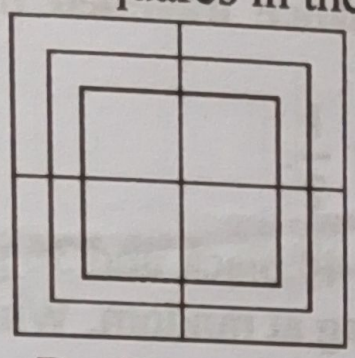
But the 2nd condition is not our interest ($\because x > 0$)

$\Rightarrow x > 0$ and $x > 1$

$\Rightarrow (0, \infty) \cap (1, \infty) = (1, \infty)$ $\quad \therefore x > 1$

Answer:- B

14. What is the number of squares in the figure given below?



A. 18 B. 12 C. 15 D. 9

Solution:- We have outer most square, middle square and inner square, and each squares are divided into four other squares. Therefore, we have a total of 15 squares.

Answer:- C

15. Assume that Ujulu saves ¼ of his income and his father had been saving half of his income. If the income of the father, which is 5000 Birr, is double of his son, how much is saved by his son?

A. 500 B. 625 C. 1250 D. 2500

Solution:

$$\text{Ujulu's save} = \frac{1}{4}\text{Ujulu's income}$$

$$\text{Father's save} = \frac{1}{2}.\text{Father's income}$$

$$= \frac{1}{2} \times 5000 = 2500 = \text{Ujulu's Income.} \quad (\because \text{FI} = 2\text{UI})$$

$$\text{Ujulu's save} = \frac{1}{4}\text{Ujulu's Income} = \frac{1}{4} \times 2500 = 625\,\text{Birr}$$

Answer:- B

16. It takes two and half an hour to get from Addis Ababa to Adama travelling at a constant rate of speed. What part of the distance is travelled by 4/3 of an hour approximately?

A. ½ B. 1/3 C. 2/3 D. ¾

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Solution:

$$V=\frac{d}{t}=\frac{d_1}{t_1} \Rightarrow dt_1=td_1 \qquad \Rightarrow d_1=\frac{dt_1}{t}=\frac{\frac{4}{3}d}{5/2}=\frac{8}{15}d.$$

$\therefore$ The part of the distance travelled by $\frac{4}{3}$ of an hour is $\frac{8}{15}$ which is approximately $\frac{1}{2}$.

Answer: A

17. A bag contains 28 yellow, 44 black and 36 green marbles. Ali takes out one marble from the bag at random. What is the probability that he will take out a green marble?

A. 1/9 B. 36% C. 1/36 D. 1/3

Solution:

Sample space $= \{28y, 44B, 36G\}$

$$\Rightarrow n(S)=28+44+36=108.$$

$$P(G)=\frac{n(G)}{n(S)}=\frac{36}{108}=\frac{1}{3}$$

Answer:- D

Quantitative Comparison

Directions: Questions 18-25 consist of two quantities, one in column A and one in column B. Compare the two quantities and answer the questions by choosing the alternative that best explains the relationship between "Column A" and " Column B". Give the answer by blackening the letter of your choice on the separate answer sheet provided.

18. Compare the two quantities.

Quantity A	Quantity B
$\frac{5}{3}$	$\frac{13}{8}$

A. Quantity A is greater than Quantity B.
B. Quantity A is less than Quantity B.
C. Quantity A is equal to Quantity B.
D. The relationship cannot be determined based on the given information.

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Solution:

Quantity A	Quantity B
$\frac{5}{3}$	$\frac{13}{8}$
$\frac{5}{3}\times\frac{8}{8}$	$\frac{13}{8}\times\frac{3}{3}$
$\frac{40}{24}$	$\frac{39}{24}$

Compare the numerators: $40 > 39$. Therefore, $Q_A > Q_B$.

Answer:- A

19. Suppose that $0 < y < 5$. Compare the following two quantities

Quantity A	Quantity B
Y^2	Y^3

A. Quantity A is greater than Quantity B.
B. Quantity A is less than Quantity B.
C. Quantity A is equal to Quantity B.
D. The relationship cannot be determined from the information given.

Solution:

Given: $0 < Y < 5$ and $\frac{Q_A}{Y^2}$ $\frac{Q_B}{Y^3}$

Let $0 < y < 1$. Then $y^2 > y^3$. Therefore, $Q_A < Q_B$.
Let $y = 1$, Then $y^2 = y^3$. Therefore, $Q_A = Q_B$.
Let $1 < y < 5$, then $y^2 < y^3$. Therefore, $Q_A > Q_B$
So, in general the relation between Q_A and Q_B cannot be determined from the given information.

Answer:- D

20. Compare the two quantities.

Quantity A	Quantity B
The percent increase from 10 meters to 15 meters	The percent increase from 50 metres to 65 metres

A. Quantity A is greater than Quantity B.
B. Quantity A is less than Quantity B.
C. Quantity A is equal to Quantity B.
D. The relationship cannot be determined from the information given.

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Solution:

Q_A	Q_B
Percent Increase : $\frac{15-10}{10}\times 100\%$	$\frac{65-50}{50}\times 100\%$
$\frac{5}{10}\times 100\%$	$\frac{15}{50}\times 100\%$ 30%
50%	

Answer:- A

Thus, 50% > 30%

Data Interpretation

Questions 56-60 are based on the hypothetical data in the table below. It is about the total population, primary school age population and School Enrolment by sex of the seven Regions in Ethiopia

Regions	Total Population		School Age Population		School Enrolment	
	Male	Female	Boys	Girls	Boys	Girls
A	1,387,000	1,500,000	162,000	166,000	220,000	267,000
B	756,000	603,000	141,000	114,000	34,000	22,000
C	9,309,000	9,317,000	1,913,000	1,887,000	1,711,000	1,572,000
D	12,883,000	12,934,000	2,709,000	2,670,000	2,734,000	2,099,000
E	7,203,000	7,287,000	1,522,000	1,500,000	1,474,000	1,113,000
F	2,268,000	1,950,000	445,000	389,000	157,000	95,000
G	2,080,000	2,143,000	443,000	431,000	447,000	436,000
Total	35,886,000	35,734,000	7,335,000	7,157,000	6.777,000	5,604,000

21. Total male population, in million, in the seven Regions in nearly.

A. 20 B. 13 C. 35 D. 26

Solution:- Given total male population = 35,886,000

In million, this nearly 35 million. **Answer: C**

22. The three least populous Regions in order are:

A. B, A, F B. B, A, E C. A, G, F D. A, D, G

Solution:- In each regions, the last digits are zero's. Omit these zero's and apply the required information for the remaining figures.

Thus, (756 + 603) < (1,387 + 1,500) < (2,268 + 1,950)

Therefore, the three least populous regions in order are region B, region A and region F **Answer:- A**

23. The proportion of girls enrolled in schools in Region D to the female population is

A. 1:2 B.1:17 C. 2: 13 D. 5:14

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Solution:

Region D:

Girls :	Female
Enrolled	population
2,099 :	12,934
≈ 2,000 :	13, 000= 2: 13

Answer: C

24. Which Region is least in enrolling girls to school compared to the boys?

A. C B. D C. E D. F

Solution:- Comparing no of girls to no of boys in each regions is time consuming. So, to tackle the problem, compare the regions given on the multiple choice.

Require C, G : B = 1,572 : 1,711 ≈ 1:1

Region D , G : B = 2,099 : 2,734 ≈ 2: 3

Region E, G:B = 1,113 : 1,474 ≈ 1 : 1

Region F , G : B = 95: 157 ≈ 100:200 = 1:2.

Therefore, Region F has least enrolling girls than boys. **Answer: D**

25. Assuming that the total population of the country at time the above data is taken 73 million, what is the share, in percent, of the population in C and D together?

A. 40 B. 50 C. 60 D. 70

Solution:

Given: Total population = 73 million.

Population in C ≈ 19 million

Population in D ≈ 26 million.

Thus, population in C and D together = 45 million.

Therefore, in percent = $\frac{45}{73} \times 100\% \approx 60\%$ **Answer: C**

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Antonyms Verbal Reasoning (1-13)

Directions: Question **1-3** consist of words printed in bold letters, followed by four other words. Choose the word that is most nearly **similar** is meaning to the word in capital letters.

1. He has reached across the world, **fomenting** violence against the west.

   A. subduing C. instigating
   B. negating D. planning

   Clues: foment: instigate, provoke, incite, Its opposite is halt, prevent, stop, negate

   Answer: B

2. My sister came back find her house in **shambles**.

   A. neat B. disorder C. mess D. ruins

   Clues: (shambles): disorder, mess, ruins

   Answer: A

3. Which of the following words cannot go together with the others?

   A. Rain B. Snow C. Wind D. Sky

   Clues: The former three refer to whether condition.

Analytical and Logical Reasoning (4 - 13)

Answer: D

4. Which two statements prove that "Lemlem and Samuel are married"?

   A. Lemlem is married to Alemu's only brother.
   B. Lemlem married her high school sweet heart.
   C. Samuel and Lemlem love each other
   D. Alemu has the same parents as Samuel.

   A. C and D B. A and C C. B and C D. A and D

   Answer: C

5. Which two statements prove that "Senait is a child"?

   A. Senait lives with her parents.
   B. Senait is the little sister of Bekele.
   C. Bekele has just turned 13.
   D. Bekele is still in school.

   A. B and D B. B and C C. C and D D. A and C

   Answer: B

6. Which two statements prove that Eleni is a doctor?

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A. Eleni loves people.
B. Eleni's father is teacher.
C. Eleni's brother is a doctor.
D. Eleni has the same job as her brother.

A. B and C B. A and C C. C and D D. A and D

Answer: C

7. Which two statements prove that Hana is younger?
A. Hana lives with her brother.
B. Hana is the little sister of Alemu.
C. Alemu is still a student.
D. Alemu has just turned 14.

A. B and D B. B and C C. C and D D. A and C

Answer: A

8. Which statements prove that "Mulualem is a lawyer"?
A. Mulualem lives with her parents.
B. Mulualem's mother is an accountant.
C. Mululem's sister is a lawyer.
D. Mulualem has the same job as her brother.

A. B and C B. A and C C. C and D D. A and D

Answer: C

9. Read the following three statements which are followed by four conclusions. Choose the conclusions which logically follow from the given.

Statements: crows are black. All birds are loud. All crows are birds.

Therefore

1. All birds black
2. All crows are black
3. All birds are loud
4. All crows are loud

A. Only (2) and (4)
B. Only (3) and (4)
C. Only (2) and (3)
D. Only (1) and (4)

Answer: A

10. **Statements:** Some questions are answers. Some answers are writers. All the writers are poets.

Conclusions:
1. Some writers are answers.
2. Some poets are questions.
3. All the questions are poets.
4. Some poets are answers

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A. Only (2) and (3)
B. Only (3) and (4)
C. Only (1) and (4)
D. Only (1) and (2)

Answer: C

11. **Statements:** Some envelops are gums. Some gums are seals. Some seals are adhesives.

Conclusion

1. Some envelopes are seals.
2. Some gums are seals.
3. Some adhesives are seals.
4. Some adhesives are gums.

A. Only 4 B. Only 2 C. Only 2 D. Only 3

Answer: C

12. Mike finished ahead of Paulos. Paulos and Berhanu both finished before Lema. Orion did not finished last. Who was the last to finish?

A. Paulos B. Berhanu C. Lema D. Orion

Answer: C

13. At university, three girls,Almaz, Sophia, and Samra, are dating three boys, Micheal, Thomas, and Maru, Almaz is dating Michael. Sophia is not dating the physics major. Samra is not dating the theatre major. Michael is a biology major. Maru is not majoring in theatre. Which of the following statements is correct, based on the above information?

A. Only Almaz and Sophia are dating two of the students.
B. Almaz is dating a theatre major.
C. Samra is dating Maru, who is majoring theatre
D. Samra is dating Maru, who is a physics major.

Answer: A

Reading Comprehension (14-21)

Reading Passage II

Marine Curie was one of the most accomplished scientists in history. Together with her husband, Pierre, she discovered radium, an element widely used for treating cancer, and studied uranium and other radioactive substances. Pierre and Marie's amicable collaboration later helped to unlock the secrets of the atom.

Marie was born in 1867 in Warsaw, Poland, where her father was a professor of physics. At an early age, she displayed a brilliant mind and a blithe personality. Her great exuberance for learning prompted her to continue with her studies after high school. She became disgruntled, however, when she learned that the university in Warsaw was closed to women.

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Determined to receive a higher education, she defiantly left Poland and in 1891 entered the Sorbonne, a French university, where she earned her master's degree and doctorate in physics.

Marie was fortunate to have studied at the Sorbonne with some of the greatest scientists of her day, one of was Pierre Curie. Marie and Pierre were married in 1885 and spent may productive years working together in the physics laboratory. A short time after they discovered radium, Pierre was killed by a horse-drawn wagon in 1906. Marine was stunned by thus horrible misfortune and endured heartbreaking anguish. Despondently she recalled their close relationship and the had shared in scientific research. The fact that she had two young daughters to raise by herself greatly increased her distress.

Curie's feeling of desolation finally began to fade when was asked to succeed her husband as a physics professor at the Sorbonne. She was we the first woman to be given a professorship at the world-famous university. In 1911 she received the Nobel Prize in chemistry for isolating radium. Although Marine Curie eventually suffered a fatal illness from her long exposure to radium, she never became disillusion about her work. Regardless of the consequences, she had dedicated herself to science and to revealing the mysteries of the physical world.

Adapted from Test Prep Review: A source from Mometrix

14. When she learned that she could not attend the university in Warsaw, she felt_____.

A. worried B. delighted C. annoyed D. hopeless

Answer: A

15. Marie _____ by leaving Poland and traveling to France to enter the Sorbonne.

A. was distressed C. showed intelligence
B. behaved D. challenged authority

Answer: C

16. _____ she remembered their job together.

A. Happily C. Worriedly
B. Tearfully D. Dejectedly

Answer: D

17. Her _____ began to fade when she returned to she Sorbonne to succeed her husband.

A. disappointment C. anger
B. wretchedness D. misfortune

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Answer: A

Reading Passage I

Regular physical activity provided numerous health benefits.: from leaner bodies and, lower blood pressure to improved mental health and cognitive functioning. As the school physical education program promotes physical activity and can teach skills as well as form or change behavior, it holds an important key to influencing health and well being across life span. To improve the fitness of students,we need to rethink the design and delivery of school-based physical education program. Adults in some countries think that information about health was more important for students to learn than content in language arts, mathematics, science, history or nay other subject. Despite this high ranking, most skills devote minimum curriculum time to teaching students how to lead healthy leaves. Our fist step might be to consider ways to increase curriculum time devoted to physical education. In addition, schools to thoughtfully analyze the design and delivery of school physical education programs to ensure that they are engaging, developmentally appropriate, inclusive and instructionally powerful.

Adopted from Test Prep Review: A resource from Momertix

18. According to this passage, regular physical activity is needed to
    A. improve ones' physical as well as metal health.
    B. improve ones' cognitive skills.
    C. lose weight.
    D. control one's blood pressure.

Answer: A

19. In order to tone up the physical education program,
    A. the program should be reoriented and implemented.
    B. a committee should be set up in every school.
    C. an assessment of the existing program should be made.
    D. it should be compulsory at school.

Answer: A

20. According to some countries, health education is more important than teaching
    A. liberal arts
    B. any subject
    C. natural sciences
    D. social sciences

Answer: B

21. The author wants the reoriented physical education program to be
    A. thoughtful
    B. relevant

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C. comprehensive D. short

Answer: C

Sentence Completion (22 -25)

22. Which continuation is light?

I don't trust social medians._____.

A. Never can I and never I will
B. Never I have and never I will
C. I have never and I will never
D. I never have and I never will

Answer: B

23. If I were in your position, I_____.

A. Would have done such silly things.
B. would have do such silly things.
C. Will never have done such silly things.
D. Would never have done such silly things.

Answer: B

24. She contracted **COVID** 19 easily because she was obese and had other pre existing diseases. The doctor said that if she _____ obese, her chances of getting the diseases would have been slim.

A. were not
B. were
C. Had been
D. had not been

Answer: D

25. I: Haven't you seen this film?

_____.

A. Yes, have I
B. Yes, I haven't
C. No, I have
D. Yes, I have

Answer: D

Analogy (26 -32)

26. EAGLE is to BIRD as ROSE is to _____

A. FLOWER
B. ANIMAL
C. BEAUTIFUL
D. GENTLEMAN

An eagle is a kind of bird as rose is a kind of flower.

Answer: A

27. CAR is to TRANSPORTATION as GAME is to _____

A. FOOTBALL
B. MANIPULATION
C. INFORMATION
D. ENTERTAINMENT

Car is for transportation as game is for entertainment

Answer: D

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28. SPEED is to FAST as TEMPERATURE is to _____
A. CLIMATE
B. WARM
C. ICE
D. THERMOMETER
Speed = fast or slow; temp = warm or cold.
Answer: B

29. FAST is to RUNNING as SLOW is to_____
A. WALKING
B. SPEED
C. LAZY
D. PACE
Answer: A

30. CANDLE: WAX
A. PAPER: PULP
B. BAMBOO: FURNITURE
C. TREE: LEAVES
D. BRANCH: WOOD
Candle is made of wax as paper is made of pulp.
Answer: A

31. CHEF: FOOD
A. BRICK: CARPENTER
B. SCULPTOR: STONE
C. BRUSH: WALL
D. PAINT: PAINTER
Chef is a professional cook (food maker) as sculptor is an artists who produces sculptures from stone.
Answer: B

32. BOARD: TRAIN
A. MOUNT: HORSE
B. BICYCLE: DRIVE
C. WALK: SHOE
D. CHAIR: SIT
Answer: D

Synonyms (33 -35)

33. The man **vehemently** denied all the charges of killing that were brought against him.
A. devoutly
B. carelessly
C. hysterically
D. forcefully
Answer: D

34. Which word means the same as "**counterfeit**?"
A. copy B. match C. robbery D. fake
Answer: D

35. He inherited a large amount of fortune from his parents. So, he lives a very **lavish** lifestyle.
A. spartan
B. humble
C. stingy
D. extravagant
Answer: D

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36. The Triangular Number Sequence T_n, can be defined recursively as $T_1 = 1$, T_n, $= T_n - 1 + n$ for $n > 1$.

What is the eleventh term of the sequence?

A. 66 B. 45 C. 78 D. 55

Solution: $T_1 = 1$

$T_n = T_{n-1} + n$, for $n > 1$

Then, $T_2 = T_{2-1} + 2 = T_1 + 2 = 1 + 2 = 3$

$T_3 = T_{3-1} + 3 = T_2 + T_3 = 3 + 3 = 6$

$T_4 = T_3 + 4 = 6 + 4 = 10$

$T_5 = T_4 + 5 = 10 + 5 = 15$

$T_6 = T_5 + 6 = 15 + 6 = 21$

$T_7 = T_6 + 7 = 21 + 7 = 28$

$T_8 = T_7 + 8 = 28 + 8 = 36$

$T_9 = T_8 + 9 = 36 + 9 = 45$

$T_{10} = T_9 + 10 = 45 + 10 = 55$

$T_{11} = T_{10} + 11 = 55 + 11 = 66$

Answer: A

37. The next term sequence 162, 54, 18, 6,... equal to

A. 2 B. 1 C. 1/3 D. 3

Solution: 162, 54, 18, 6, x

since $\frac{54}{162} = \frac{18}{54} = \frac{1}{3}$, then $\frac{x}{6} = \frac{1}{3}$

$\Rightarrow \quad x = 2$

Answer: A

38. The first four terms of an arithmetic sequence are 39,33, 27 and 21.

A. –21 B. –15 C. –33 D. –27

Solution: 39, 33, 27, 21, ---- ← AP with common difference $d = -6$

Then, $A_{10} = A_1 + 9d$

$= 39 + 9(-6)$

$= 39 - 54 = -15$

Answer: B

39. The sum of the first n terms of an arithmetic progression whose first term is $2n^2 + n$ then what is the tenth term of the sequence?

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A. 39 B. 40 C. 38 D. 41

Solution: $S_n = 2n^2 + n$, Then

$$A_{10} = S_{10} - S_9$$
$$= (2(10^2) + 10) - (2(9^2) + 9)$$
$$= 2(100) + 10 - (2(81) + 9)$$
$$= 210 - 171 = 39$$

Answer: A

For Questions 40 - 45 Use The Following Information:

The following table gives the expenditure (in millions of Birr) of a certain company per year over five years.

Year	Items of Expenditure				
	Salary	Fuel and Transport	Bonuses	Interest on Loans	Taxes
2008	288	98	3.1	23.4	83
2009	342	112	2.52	32.5	108
2010	324	101	3.84	41.2	74
2011	336	133	3.68	36.4	88
2012	422	142	3.96	49.4	98

40. What is the average amount of interest on loans per year that the company has to pay during the five year period?

A. 33.72 B. 33.43 C. 36.58 D. 34.18

Year	Items of Expenditure					
	Salary	Fuel and Transport	Bonuses	Interest on Loans	Taxes	Total
2008	288	98	3.1	23.4	83	495.5
2009	342	112	2.52	32.5	108	597.02
2010	324	101	3.84	41.2	74	544.04
2011	336	133	3.68	36.4	88	697.08
2012	422	142	3.96	49.4	98	715.36
Total	1712	586	17.1	182.9	451	

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Solution: Average amount of interest on loans $= \frac{182.9}{5} = 36.58$

Answer: C

41. What percent of the total amount of salary was paid as bonuses by the company during the given period?

A. 0.5% B. 0.1% C. 1.25% D. 1.0%

Solution:

$$\%\text{ of bonuses} = \frac{\sum \text{bonuses}}{\sum \text{salary}} \times 100\%$$

$$= \frac{17.1}{1712} \times 100\%$$

$$= 0.009988 \times 100\%$$

$$= 0.9988\% \approx 1\%$$

Answer: D

42. In which year did the company have the highest expenditure?

A. 2009 B. 2008 C. 2010 D. 2012

The highest expenditure = 715.36 which is in 2012.

Answer: D

43. What is the total expenditure of the company in the year 2012?

A. 715.36 B. 1710 C. 456.36 D. 586

Total expenditure in 2012 = 715. 36

Answer: A

44. For what item or purpose was the company's expenditure the least during the given period?

A. Bonus C. Taxes

B. Salary D. Transport and Fuel

The least expenditure during the given period in Bonus.

Answer: A

45. If $x = 15\%$ of 30 and $y = 30\%$ of 15, then which of the following is true?

A. x and y are equal

B. x is less than y

C. The relation between x and y cannot be determined

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D. y is less than x

Solution

$x = 15\% \times 30$, $y = 30\% \times 15$

$= \frac{15}{10\cancel{0}} \times 3\cancel{0}$ $= \frac{3\cancel{0}}{10\cancel{0}} \times 15$

$= \frac{45}{10}$ $= \frac{45}{10}$

$= 4.5$ $= 4.5$

$\therefore \quad x = y$

Answer: A

46. Student X and Y have 200 Birr. If $\frac{4}{15}$ of X's share is equal to $\frac{2}{5}$ of Y's share, how much amount does Y have?

A. 800 Birr B. 1500 Birr C. 1200 Birr D. 1100 Birr

Solution

$x + y = 2000$ and $\frac{4}{15}x = \frac{2}{5}y$

$\Rightarrow \frac{2}{3}x = y$

So, $x + \frac{2}{3}x = 2000$

$\Rightarrow \frac{5}{3}x = 2000$

$\Rightarrow x = 1200\text{Birr}$

$\Rightarrow y = \frac{2}{3} \times 1200 = 800\text{Birr}$

Answer: A

47. What is the simplified form of $(2x + 3)(x + 6) - (2x - 5)(x + 10)$?

A. 16 B. 32 C. 72 D. 68

Solution: $(2x + 3)(x + 6) - (2x - 5)(x + 10)$

$= 2x^2 + 12x + 3x + 18 - (2x^2 + 20x - 5x - 50)$

$= 2x^2 + 15x + 18 - 2x^2 - 15x + 50 = 68$

Answer: D

48. If the square of a number increased by two times the number equal to six times the number increased by twelve, which one of the fol-

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lowing is the possible value of the number?

A. 4 B. 6 C. –4 D. –6

Solution: Let x be the number. Then

$$x^2 + 2x = 6x + 12$$
$$\Rightarrow x^2 - 4x - 12 = 0$$
$$\Rightarrow (x-6)(x+2) = 0$$
$$\Rightarrow x - 6 = 0 \text{ or } x + 2 = 0$$
$$\Rightarrow x = 6 \text{ or } x = -2$$

Answer: B

49. The monthly telephone bill of Getachew was Birr 400 and it increases by 10% after 12 months, and another 20% increase is applied six months later. What is the price of Getachew's telephone bill after 18 months?

A. 520 Birr B. 528 Birr C. 489 Birr D. 430 Birr

Solution: Initial amount = Birr 400

After 12 months = Birr 400 + 10% × 400
= Birr 400 + Birr 40
= Birr 440

After 18 months = Birr 440 + 20% × 440
= Birr 440 + Birr 88
= Birr 528

Answer: B

50. The sum of the ages of the father and his son is equal to sixty years. Six years ago the age of the father was five times the age of the son. What will be the be of the son six years from now?

A. 19 years B. 23 years C. 22 years D. 20 years

Solution: Let F and S are the present age of the father and his son

resp. $F + S = 60$ --- *

$F - 6 = 5(S - 6)$

$\Rightarrow F - 6 = 5S - 30$

$\Rightarrow F - 5S = -24$ ----**

Solve, equations (*/ and **) simultaneously.

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$$-\begin{cases} F+S=60 \\ F-5S=-24 \end{cases}$$

$$(S(-5S))=60-(-24)$$

$$6S=84$$

$$S=\frac{84}{6}=14$$

The age of the son after 6 years will be $14+6=20$ years.

Answer: D

51. What is the sum of the following infinite series

$$S=2-\frac{1}{3}+1-\frac{1}{9}+\frac{1}{2}-\frac{1}{27}+\frac{1}{4}-\frac{1}{81}...?$$

A. 7/2 B. 32/9 C. 41/9 D. 9/2

Solution

$$S=2-\frac{1}{3}+1-\frac{1}{9}+\frac{1}{2}-\frac{1}{27}+\frac{1}{4}-\frac{1}{81}+...?$$

$$=\underbrace{\left(2+1+\frac{1}{2}+\frac{1}{4}+---\right)}_{\text{Infinite Geometric}}-\underbrace{\left(\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\frac{1}{81}+---\right)}_{\text{Infinite Geometric series with r=1/3}}$$

$$=\left(\frac{G_1}{1-R}\right)-\left(\frac{G_1}{1-r}\right)$$

$$=\frac{2}{1-1/2}-\frac{\frac{1}{3}}{1-1/3}=4-1/2=7/2$$

Answer: A

52. If $(1^2+2^2+3^2+...+10^2)=a$ then what is value of $(2^2+4^2+6^2+...+20^2)$ in terms of a?

A. 4a B. 2a C. a+4 D. a+2

Solution: Given $1^2+2^2+3^2...+10^2=a$

$$2^2+4^2+6^2+...+20^2=(1\times2)^2+(2\times2)^2+(2\times3)^2+...+(2\times10)^2$$

$$=1^2\times2^2+2^2\times2^2+2^2\times3^2+...+2^2\times10^2$$

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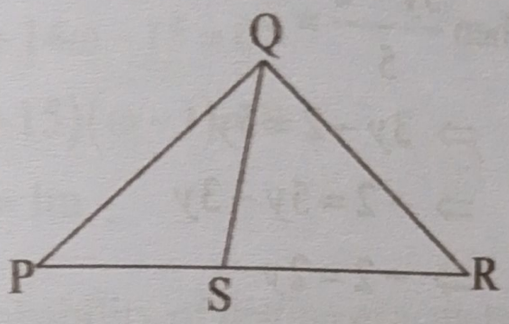
$$= 2^2\left[1^2 + 2^2 + 3^2 + \ldots + 10^2\right] = 4a$$

Answer: A

53. Consider the figure given below.



It is given that PQ = RQ. Compare the following two quantities X and Y information are given below.

Quantity X	Quantity Y
PS	SR

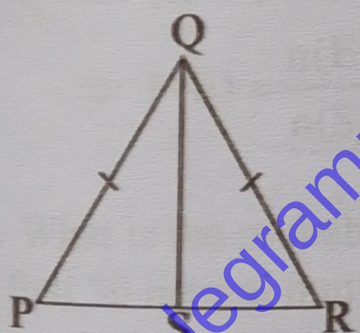
A. Quantity Y is greater than Quantity X.

B. Quantity X is greater than Quantity Y.

C. The relation cannot be determined with the information provided.

D. The two quantities are equal.

Solution



ΔPQR is an isosceles triangle

If $\overline{QS}$ bisects $\angle Q$, then PS = SR.

Otherwise the following relation holds true.

PS > SR or PS < SR or PS = SR

∴ The relation cannot be determined with the information provided

Answer: C

54. If y > 2, compare the following two quantities X and Y whose information are given below.

Quantity X	Quantity Y
$\frac{3y-2}{5}$	y

A. Quantity Y is greater than Quantity X.

B. Quantity X is greater than Quantity Y.

C. The relation cannot be determined with the information provided.

D. The two quantities are equal.

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Solution: Given $y > 2$ and $x = \frac{3y-2}{5}$

Assume : $x = y$., Then $\frac{3y-2}{5} = y$

$\Rightarrow 3y - 2 = 5y$

$\Rightarrow -2 = 5y - 3y$

$\Rightarrow -2 = 2y$

$\Rightarrow y = -1$ which is not.

Assume $x > y$. Then $\frac{3y-2}{5} > y$

$\Rightarrow 3y - 2 > 5y$

$\Rightarrow 3y - 5y > 2$

$\Rightarrow -2y > 2$

$\Rightarrow y < -1$ which is not.

Assume $x < y$. Then $\frac{3y-2}{5} < y$

$\Rightarrow 3y - 2 < 5y$

$\Rightarrow 3y - 5y < 2$

$\Rightarrow -2y < 2$

$\Rightarrow y > -1$

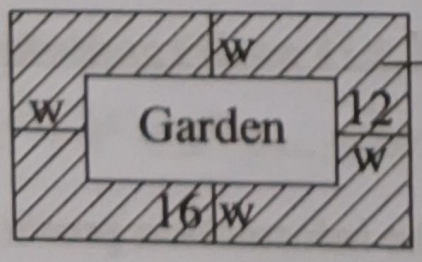
$\Rightarrow y > 2$

Answer: A

55. A rectangular garden measuring 12 meter by 16 meter is to have a pedestrian pathway that is w meters wide installed all way around so that the total area including the pedestrian path way is 252 m^2. What is the width of the pathway?

A. 12m B. 1.5m C. 1.2m D. 15m

Solution



pedestrian since $A_{garden} = 192m^2$ is smaller than $A_{total} = 252m^2$ the garden is in the inferior.

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$$A_{total} = (16 + 2\omega)(12 + 2\omega)$$

$$252 = 192 + 32\omega + 4\omega^2 + 24\omega$$

$$\Rightarrow 4\omega^2 + 56\omega - 60 = 0$$

$$\Rightarrow \omega^2 + 14\omega - 15 = 0$$

$$\Rightarrow (\omega + 15)(\omega - 1) = 0$$

$$\Rightarrow \omega = 1m$$

No answer

56. What is the probability of getting exactly two tails when three coins are tossed at the same time?

A. $\frac{1}{4}$ B. $\frac{1}{8}$ C. $\frac{3}{4}$ D. $\frac{3}{8}$

Solution

Tossing three coins at the same time is the same as to tossing a coin three times.

So, the following three diagram represents the experiments.

Then, S = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT} and

E = {HTT, THT, TTH}

$$\Rightarrow P(E) = \frac{n(E)}{n(S)} = \frac{3}{8}$$

Answer: D

57. What is the probability that a number selected at random from the first 50 natural numbers is prime and odd number?

A. $\frac{3}{10}$ B. $\frac{7}{25}$ C. $\frac{8}{25}$ D. $\frac{3}{5}$

Solution: n (S) = 50

E = {3, 5, 7, 11, 13, 17, 23, 29, 31, 37, 41, 43, 47}

$$\Rightarrow p(E) = \frac{n(E)}{n(S)} = \frac{14}{50} = \frac{7}{25}$$

Answer: B

58. If $3x + 7y = 80$ and $x - y = 10$, then what is the value of $x + y$?

A. 10 B. 15 C. 25 D. 20

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Solution

$$3\begin{cases}3x+7y=80\\x-y=10\end{cases}=\frac{\begin{cases}3x+7y=80\\3x-3y=30\end{cases}}{10y=50}$$

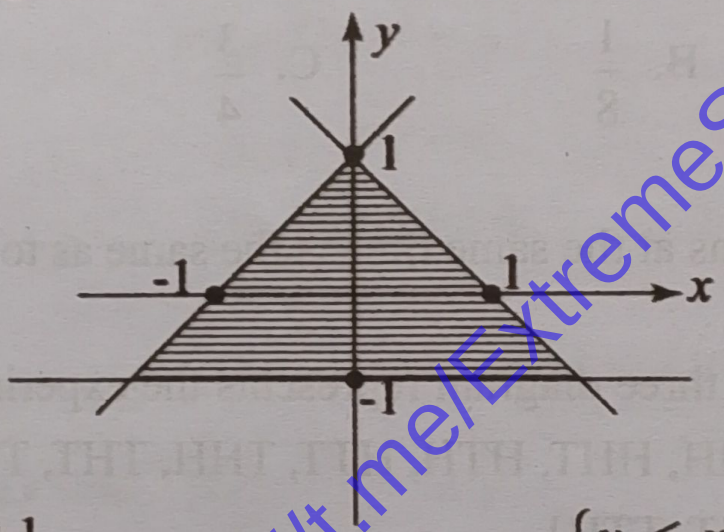
$y=5$ and $x-y=10$

$\Rightarrow x=15$

$\therefore\ x+y=5+15=20$

Answer: D

59. Which of the following system of linear inequalities is represented by the graph given below?



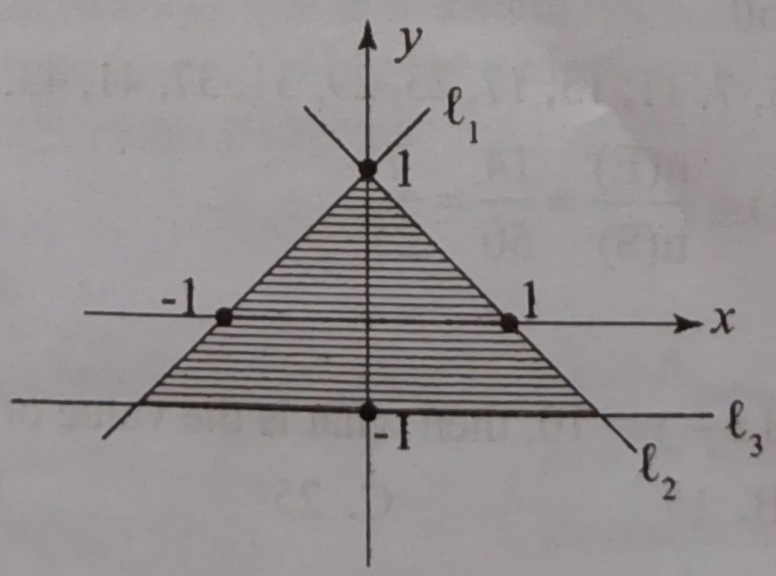
A. $\begin{cases}y\le x+1\\y\le x-1\\y\le -1\end{cases}$

B. $\begin{cases}y\le x+1\\y\le -x+1\\y\le -1\end{cases}$

C. $\begin{cases}y\le x+1\\y\le x-1\\y\ge -1\end{cases}$

D. $\begin{cases}y\le x+1\\y\le -x+1\\y\ge -1\end{cases}$

Solution



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Equation of $\ell_1 : y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$\Rightarrow y - 1 = \frac{0-1}{-1-0}(x-0)$

$\Rightarrow y - 1 = x$

$\Rightarrow y = x + 1$

Equation of $\ell_2 : y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$\Rightarrow y - 1 = \frac{1-0}{0-1}(x-1)$

$\Rightarrow y = -1(x-1)$

$\Rightarrow y = -x + 1$

Equation $\ell_3 : y = -1$

Now, the region determined by l_1, l_2 and l_3 is $y \le x + 1, y \le -x + 1$ and $y \ge -1$

Answer: D

60. The square of the 7th term of an arithmetic progression with positive common difference is equal to the product of the 3rd and 17th terms, then what is the ratio of the first term to the common difference?

A. 3 : 4 B. 2 : 3 C. 4 : 3 D. 3 : 2

Solution:

$A_7^2 = A_3 \cdot A_{17}$

$\Rightarrow (A_1 + 6d)^2 = (A_1 + 2d)(A_1 + 16d)$

$\Rightarrow A_1^2 + 12A_1d + 36d^2 = A_1^2 + 16A_1d + 2A_1d + 32d^2$

$\Rightarrow 12A_1d + 36d^2 = 18A_1d + 32d^2$

$\Rightarrow 36d^2 - 32d^2 = 18A_1d - 12A_1d$

$\Rightarrow 4d^2 = 6A_1d$

$\Rightarrow 2d = 3A_1$

Then, $A_1 : d = A_1 : \frac{3}{2}A_1 = 1 : \frac{3}{2} = 2 : 3$

Answer: B

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SECTION ONE: Verbal Reasoning (Questions 1-35)

This section which is subdivided into analogy, synonyms, classification, reading comprehension, analytical and logical reasoning contains 35 verbal reasoning items altogether. Each subsection is followed by its specific direction.

Synonyms (1-3)

Directions: Questions **1-3** consist of words printed in capital letters followed by four words. Choose the word that is **most nearly similar** in meaning to the word in capital letter. Then **blacken** the letter of your choice on the separate answer sheet provided.

1. ATTACHMENT
   A. affinity
   B. influence
   C. causation
   D. appendage

   Clues: (N) is to mean something connected to or joined to the machine or implement.

   Answer: D

2. FOUND
   A. see B. establish C. realise D. search

   Clues: (V) to establish; to set up; to institute

   Answer: B

3. ADVICE
   A. council B. counsel C. practice D. proposal

   Clues: counsel (v) which means advise
   council (N) which means leader
   (example: students council, city council)

   Answer: B

Antonyms (4-6)

Directions: Question **4-6** consist of words printed in capital letters followed by four words. Choose the word that is **most nearly opposite** in meaning to the word in capital letter. Then **blacken** the letter of your choice on the separate answer sheet provided.

4. IMPARTIAL
   A. hostile B. need C. biased D. worried

   Clues: (adj): = fair, equal, equal treatment ≠ biased/ unfair/ discriminate

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Answer: C

5. PROGRESS

A. silence B. movement C. reversion D. advance

Clues: (N) = movement, advancement ≠ regress/ retrogress

Answer: C

6. CANDID

A. vague B. rude C. cunning D. deceptive

Clues: (adj) = movement, advancement ≠ deceptive, dissembling, uncandid, fallacious.

Answer: D

Analogy (7-13)

Direction: To answer the items **7-13**, read each of the following words carefully and choose the word that correctly completes the analogy. Then **blacken** the letter of your choice on the separate answer sheet provided.

7. BRIDGE is to RIVER as TUNNEL is to _____.

A. nails C. mountain

B. underway D. rock

Clues: Bridge is used to cross the river on as tunnel is used to pass through under water/ ground.

Answer: B

8. PROSE is to a POETRY as CONVERSATION is to _____.

A. song B. poem C. language D. listening

Clues: prose (ለድ ፅሁፍ ዝርው ፅሁፍ) is opposite for poetry (ሥነ ግጥም) prose is known by its greater irregularity and variety of rhythm; it is also advisor ordinary style and quality

Poetry is a very beautiful or graceful song or verse.

As: conversation is a dialogue (informal talk) between two or more people; while listening is the way of perceiving sound.

Answer: D

9. READING is to a KNOWLEDGE as WORK is to _____

A. money C. experience

B. employment D. engagement

Clues: We read to gain knowledge as we work to gain money.

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Answer: A

10. COLLEGE is related to TEACHERS in the same way as HOSPITAL is related to:

A. doctors B. patients C. medicine D. beds

Clues: College is a place where teachers work as hospital is a place where doctors work.

Answer: A

11. TABLE is related to WOOD in the same way as SHIRT is related to:

A. cotton B. cloth C. dress D. uniform

Clues: Table is made of wood as shirt is made of cotton.

Answer: A

12. WALK is related RUN in the same as BREEZE is related to:

A. cold B. dust C. wind D. air

Clues: Walk is a light or gentle run, as breeze is a light/gentle wind.

Answer: C

13. WAX is related to GREASE in the same as MILK is related to:

A. drink B. ghee C. protein D. curd

Clues: Wax forms grease when melts as milk forms curd when becomes sour.

Answer: D

Classification Items (14-17)

Directions: In questions 14-17, a set of three words is given. Three of the words are related in some ways. The remaining word is not related to the rest. Choose the word which does not fit the relation and **blacken** the letter of your choice on the separate answer sheet provided.

14. A. Haughty B. Reticent C. Sociable D. Aloof

Clues: Sociable/ friendly; the rest are the opposite of it.

Answer: C

15. A. Treachery B. Morbid C. Deceit D. Swindle

Clues: Morbid(disease, unhealthy), the rest are/ have the same meaning.

Answer: B

16. A. Sporadic
B. Frequent
C. Intermittent
D. Scarce

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Clues: frequent (many times), the rest mean seldom (infrequent)

Answer: B

17. A. Cease
    B. Launch
    C. Initiate
    D. Commenece

Clues: cease (halt, stop) example: cease fire -(ተኩስ አቁም) the rest are the opposite of it.

Answer: A

Reading Comprehension (18-25)

Direction: Read the passages carefully and answer questions **18-25**. Each question will be based on the information stated or implied in the passage. Then **blacken** the letter of your choice on the separate answer sheet provided.

Reading comprehension I

Read the following passage and answer questions (18-21)

The coconut is an unusual food for many reasons. It is technically a seed, produced by the coconut palm tree, and as such is one of the largest edible seeds produced by any plant. Its unusual contents also make it unique in the seed world the interior consists of both "meat" and "water." The meat is the white pith with which we are all familiar, as it is used extensively for cooking and flavourings; the coconut water is a white liquid that is very sweet and thirst-quenching. Portuguese explores gave the nut its name in the 15th century, referring to it as coco, meaning "ghost" in their language. The three dimples and the hairy texture reminded them of a ghost's face, and the tree has retained that name ever since. The coconut has many varied uses. It is used to make margarine, as well as various cooking oils, and these cooking oils are used by fast-food restaurants around the world to make such diet staples as French fries. The coconut fluid is a favorite drink in hot climates, providing a cool and refreshing beverage right off the tree. This water is also used by manufactures of various sports drinks because of its isotonic electrolyte properties. Even the shell itself hs many uses, including cattle food and fertilizer.

Adopted from Reading Comprehension 4th edition

18. The coconut earned the nick name "ghost" because
    A. its pale color.
    B. it resembles ghost's a face.
    C. it is round.
    D. of its smell

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Answer: B

19. What is the main focus of this passage?
    A. the history of coconut
    B. Coconut trees have many uses.
    C. how cooking oil is made
    D. Portuguese discoveries.

Answer: A

20. The passage implies that coconut palms are
    A. known valuable plants.
    B. coconut oil is the best way to cook.
    C. Portuguese explorers loved coconuts.
    D. coconut palms are good shade trees.

Answer: B

21. Coconut differs from the other seeds because of its
    A. color.
    C. contents.
    B. special uses.
    D. flavour.

Answer: C

Reading comprehension II

Read the following passage and answer questions (22-25)

Dogs and cats should never be permitted to eat chocolate because chocolate works like a poison in their bodies. Chocolate contains a chemical called theobromine, which is similar to caffeine. Human bodies are able to process the theobromine without any ill side effects, but dogs and cats cannot. Different types of chocolate contain different amounts of theobromine. It would take 20 ounces of milk chocolate to kill a 20-pound dog, but only two ounces of baker's chocolate or six ounces of semi-sweet chocolate. The amounts, of course, are much smaller for a cat, whose body weight is typically less than that of a dog are. Most cats are not naturally attracted to eating chocolate, but many dogs are. Dogs by nature will sample nearly anything that they see masters eating, so pet owners must take are to keep all chocolate products well out of reach of their dogs and cats.

Adapted from Reading Comprehension 4th edition

22. According to the passage, why is chocolate poisonous for dogs and cats?

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A. It contains caffeine.

B. Chocolate is made from processed cocoa.

C. It gets stuck in their intestines.

D. They cannot process theorbromine.

Answer: D

23. According to the passage, how much milk chocolate would be poisonous to a cat?

A. substantially less than 20 ounces

B. substantially more than 20 ounces

C. approximately one pound

D. half a Hershey bar

Answer: A

24. Why might a dog eat chocolate, according to the passage?

A. Because it tastes good

B. Dogs won't eat chocolate.

C. They can smell the theobromine.

D. Dogs like to imitate their owners.

Answer: D

25. What best summarizes this passage?

A. Look before you leap.

B. Pet food for pets, people food for people.

C. Monkey see, monkey do.

D. A penny saved is penny earned.

Answer: B

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Analytical and Logical Reasoning Questions (26-35)

Direction: Question **26-35** deal with logical and analytical reasoning. Read the given set of conditions and assumption for each question. Choose the answer that you think is most appropriate among the given alternatives. Then **blacken** the letter of your choice on the separate answer sheet provided.

XYZ, a specialty restaurant, is open for business every Monday through Saturday but is closed Sundays. Lunch is the only meal served on Mondays, Tuesdays and Thursdays. Dinner is the only meal served on Wednes-

days, Fridays and Saturdays. The restaurant's floors are polished and its plants are watered only on days that XYZ is open for business, according to the following schedule:

- Plants are watered two days each week, but never on consecutive days and never on the same day are that floors polished.
- Floors are polished on Monday and two other days each week, but never on consecutive days and never on the same day that plants are watered.

26. According to the schedule, which of the following could be true?

A. Plants are watered on Saturday.

B. Lunch is served on Tuesday.

C. Floors are polished on Tuesday.

D. Dinner is served on Thursday.

Answer: A

27. According to the schedule, the restaurant's floors are polished on either

A. Tuesday or Wednesday.

B. Tuesday or Thursday.

C. Wednesday or Friday.

D. Thursday or Friday.

Answer: C

28. If dinner is served on a day that plants are watered, which of the following must be true?

A. Plants are watered on Tuesday.

B. Floors are polished on Thursday.

C. Plants are watered on Wednesday.

D. Floors are polished on Wednesday.

Answer: C

29. Assume that floors are polished on consecutive instead on non-consecutive days but that all other scheduling policies are unchanged. For how many of the six days can it be determined both whether plants watered and whether floors are polished.

A. Two B. Three C. Four D. Six

Answer: D

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Read the following text and answer the question that follow it.

A College in Addis Ababa has built its reputation for academic excellence largely on significant contributions from wealthy alumni who are fans of the colleges' football team. Although the team has won more national championships over the years than any other team in its division, this year it did not even win the division title, and so the College can expect to see a decline in alumni contribution next year.

30. The above argument relies which of the following assumptions about the college?

A. The college's reputation for academic excellence depends on the performance of its football team.

B. Contributions from alumni are needed for the college to products a winning football team.

C. The college's alumni contributions depend to an extent on a winning record by the college's football team.

D. The college's football team will continue its losing streak next year.

Answer: C

Read the following text and answer the question that follow it.

It is well known that the world urgently needs the adequate distribution of food so that everyone gets enough. Adequate distribution of medicine is just as urgent as the need for food. Medical expertise and medical supplies need to be redistributed throughout the world so that in emerging nations will have proper medical care.

31. This paragraph best supports the statement that

A. food production in emerging nations has slowed during the past several years.

B. most of the world's doctors are selfish about giving time and money products.

C. the medical-supply industry should step up production of its products.

D. many people who live in emerging nations are not receiving proper medical care.

Answer: D

32. Five children are sitting in a row. S is sitting next to P but not T. K is

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sitting next to R who is sitting on the extreme left and T is not sitting next to K. Who are sitting adjacent to S?

A. A. K & P B. R & P C. Only & P D. P & T

Answer: D

33. In a family, there are husband-wife, two sons and two daughters. All the ladies were invited to a dinner. Both sons went out to play. Husband did not return from office. Who was at home?

A. Only wife was at home.
B. Nobody was at home.
C. Only sons were at home.
D. All ladies were at home.

Answer: A

34. Read the following statement and the conclusions. Then choose the conclusion that can be reached based on the statement.

Statement: Parents are prepared to pay any price for an elite education to their children.

Conclusions: _____.

A. All parents these days are very well off
B. Parents have an obsessive passion for perfect development of their children through good schooling
C. Parents are read to do whatever they can to satisfy their children
D. Rich parents were more concerned for their children's education that poor parents are

Answer: B

35. Rearrange the following five sentences in proper sequence to from a meaningful paragraph, then answer the questions given below.

1. Kemal received a call to attend the interview.
2. He applied for a new job.
3. Kemal was an ambitious boy.
4. But, he was not happy there.
5. His father had put him in a clerical job.

A. 2,4,5,3,2, B. 4,2,1,5,3 C. 3,5,4,2,1 D. 2,5,4,3,1

Answer: C

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DIRECTIONS: Solve each of the problems given in questions **36-60** below and select the best answer from the alternatives and **blacken** the letter of your choice on the separate answer sheet.

36. Which one of the following is equal to the arithmetic sequence $(a_n)^{\infty}\ n=1$, if $a_5 + a_9 = 72$ and $a_7 + a_{12} = 97$?

A. 6, 11, 16, 21, 26, ...
C. 4, 9, 14, 19, 24, ...
B. 5, 10, 15, 20, 25, ...
D. 3, 8, 13,18, 23, ...

Solution:

$\Rightarrow \quad a_7 + a_{12} = 97$

$a_5 + 2d + a_9 + 3d = 97 \Rightarrow a_5 + a_9 + 5d = 97$

$72 + 5d = 97$ and $5d = 25$

$d = 5$

Also we have,

$a_5 + a_9 = 72$

$a_5 + a_5 + 4d = 72 \Rightarrow 2a_5 + 20 = 72$

$2a_5 = 52 \Rightarrow a_5 = 26$

Answer: A

37. Which term of the following geometric sequence 2, 8, 32, 128,... is 2048?

A. 11^{th}
B. 12^{th}
C. 6^{th}
D. 7^{th}

Solution: 2, 8, 32, 128,... 2048

$\Rightarrow \quad G_1 = 2, \qquad r = 4$

$G_n = r^{n-1}G_1 \Rightarrow 2048 = 4^{n-1} \cdot 2$ and $1024 = 4^{n-1}$

$2^{10} = 2^{2n-2} \Leftrightarrow 2n - 2 = 10,$

$n = 6, \quad i.e \ G_6 = 2048$

Answer: C

38. In the arithmetic sequence: -1, n, 13, ..., which one of the following is the value of n?

A. 7
B. 6
C. 9
D. 8

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Solution:

$-1, n, 13, \ldots \leftarrow$ AP

$n-(-1)=13-n \Rightarrow n+1=13-n$

$2n=12=6$

Answer: B

39. Which one of the following is a term in the sequence 6, 14, 22,...?

A. 246 B. 264 C. 284 D. 258

Solution: 6, 14, 22,....

$14-6=22-14=8=d$

$\Rightarrow$ It is an AP.

$\Rightarrow A_n = 4_1 + (n-1)d = 6+(n-1)8 = 8n-2$

Let $A_n = 246 \Rightarrow 8n-2=246$

$8n=248 \Rightarrow n=31$

$\therefore$ 246 is a term of the sequence.

Let $A_n = 264$, so that

$\Rightarrow 8n-2=264$

$8n=266$

$\therefore$ 264 is not a term of the sequence.

Answer: A

40. Which one of the following is equal to the series $\sum_{n=1}^{\infty}(2n-5)$?

A. $(1)+(5)+(9)+\ldots$ C. $(1)+(-3)+(-7)+\ldots$

B. $(1)+(-3)+(1)+\ldots$ D. $(-3)+(-1)+(1)+\ldots$

Solution:

$$\sum_{n=1}^{\infty}(2n-5)=(2-5)+(2\times 2-5)+(2\times 3-5)+\ldots$$

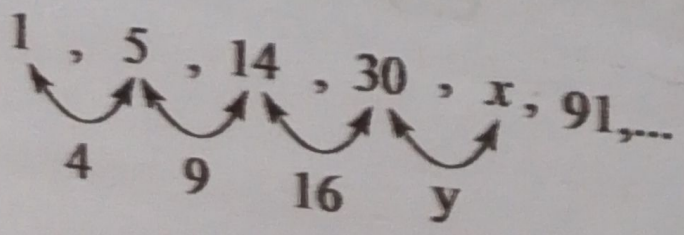
$$=-3+(-1)+1+\ldots$$

Answer: D

41. Which one of the following is the value of x in the sequence: 1, 5, 14, 30, x, 91,...?

A. 85 B. 65 C. 55 D. 75

Solution:



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$\Rightarrow 2^2, 3^2, 4^2, y = 5^2 = 25$

$\therefore x = 30 + 25 = 55$

Answer: C

42. Which of the following is the sum of the series: $3+1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+...?$

A. $\frac{9}{2}$ B. $\frac{3}{2}$ C. $\frac{2}{9}$ D. $\frac{1}{3}$

Solution:

$3+1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\dots$

– It is a geometric series with $G_1 = 3$, and $r = \frac{1}{3}$.

$\Rightarrow S^{\infty} = \frac{G_1}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{3}{2/3} = \frac{9}{2}$

Answer: A

43. Which of the following is the quadrant that the common solution of the equations $2x+3y=2$ and $3x+2y=2$ is found?

A. I quadrant. C. II quadrant.

B. III quadrant. D. IV quadrant.

Solution: Solving together by substition:

Take $3x+2y=2$ and solve for y:

$\Rightarrow 2y = -3x+2 \Leftrightarrow y = \frac{-3}{2}x+1$

- Put $y = \frac{-3}{2}x+1$ in equation $2x+3y=2$

$\Rightarrow 2x+3\left(\frac{-3}{2}x+1\right)=2$

$\Rightarrow \frac{4x-9x}{2} = 2-3 = -1$

$\Rightarrow -5x = -2 \Leftrightarrow x = \frac{2}{5}$

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• solving for $y = \frac{-3}{2}\left(\frac{2}{5}\right)+1 = \frac{2}{5}$

Intersection poin t: $\left(\frac{2}{5},\frac{2}{5}\right)$

$\therefore$ It lies in quadrant I.

Answer: A

44. Given the inequality $y \leq 2x+2$, which of the following ordered pairs of numbers is **NOT** a solution for the given inequality?

A. (2, 3) B. (0, 2) C. (–3, 3) D. (–1, 0)

Solution:

$y \leq 2x+2$

$3 \leq 2(-3)+2$

$3 \leq -6+2$

$3 \leq -4(\textit{false})$

Answer: C

45. The equation $T = 25 + 3c$ is used to model the number of chirps, c, made by a certain species of cricket in one minute, and the temperature, T, in degrees Fahrenheit. According to this model, what is the estimated increase in temperature, in degrees Fahrenheit, when the number of chirps in one minute is increased by 1?

A. 3 B. 5 C. 25 D. 28

Solution: $T = 25 + 3c$

Let $c_1 = c+1$

$\Rightarrow T_1 = 25+3c_1 = 25+3(c+1) = 25+3c+3$

$T_1 = T+3$

Answer: A

46. At a certain school, if the ratio of teachers to students is 1 to 10, which one of the following can be a total number of teachers and students in the school?

A. 100 B. 121 C. 144 D. 1,011

Solution:

T: S = 1:10 $\Rightarrow$ S = 10T

Total = n(Teacher) + n(Students) = n(T) + n(S) = T + 10T = 11T

$\Rightarrow$ The total number of teaches and students will be a multiple of 11.

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Answer: B

47. If x is 30% of 400, y is 40% of x, and z is 25% of y, which of the following is equal $x + y + z$?

A. 120 B. 180 C. 150 D. 175

Solution:

$$x = 30\% \text{ of } 400 = \frac{30}{100} \times 100 = 120$$

$$y = 40\% \text{ of } x = \frac{40}{100} \cdot (120) = 48$$

$$z = 25\% \text{ of } y = \frac{25}{100} \times 48 = 12$$

$$\Rightarrow x + y + z = 120 + 48 + 12 = 180$$

Answer: B

48. A radioactive substance decays at an annual rate of 17%. If the initial amount of the substance is 650 grams, which one of the following function $h(t)$ models the remaining amount of the substance, in grams after t years?

A. $h(t) = 0.17(650)^t$ C. $h(t) = 083(650)^t$

B. $h(t) = 650(0.17)^t$ D. $h(t) = 650(0.83)^t$

Solution:- The remaining amount of the substance after

$$1\text{year} = 650 - 17\% \cdot 650 = 650\left(1 - \frac{17}{100}\right)$$

$$2\text{years} = 650\left(1 - \frac{17}{100}\right) - \frac{17}{100}\left(650\left(1 - \frac{17}{100}\right)\right)$$

$$= 650\left(1 - \frac{17}{100}\right)\left(1 - \frac{17}{100}\right) = 650\left(1 - \frac{17}{100}\right)^2$$

$$\vdots$$

$$t \text{ years} = 650\left(1 - \frac{17}{100}\right)^t = 650\left(\frac{83}{100}\right)^t = 650(0.83)^t$$

Answer: D

49. Which of the following is equal to the expression $\left(\frac{2y+6}{4}\right) \cdot \left(\frac{3y-36}{3y+9}\right)$ for $y \neq -3$?

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A. $\frac{12y^2-126}{12y+32}$ B. $\frac{y-12}{2}$ C. $\frac{8y-30}{3y+13}$ D. $y-6$

Solution:

$$\left(\frac{2y+6}{4}\right)\cdot\left(\frac{3y-36}{3y+9}\right)=\frac{2(y+3)}{4}\cdot\frac{3(y-12)}{3(y+3)}$$

$$=\frac{y+3}{2}\cdot\frac{y-12}{y+3}=\frac{1}{2}(y-12)$$

Answer: B

DIRECTIONS: Questions **50 and 51** both consist of two quantities, one in Column A and another one in Column B. Compare the two quantities and answer the questions by choosing the alternative that best explains the relationship between "Column A" and "Column B".

50. The diameter of a circle O is the same as the length of the side of a square S.

Column A	Column B
The area of circle O	The area of square S

A. The quantity in column A is greater.
B. The quantity in column B is greater.
C. The two quantities are equal.
D. The relationship cannot be determined.

Solution:

d = diameter of a circle and S = side of a square

$$\text{Area of a circle} = \pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \frac{1}{4}\pi d^2$$

$$\Rightarrow \text{Area of circle} = \frac{1}{4}\pi d^2 = \frac{1}{4}\pi S^2$$

$$\Rightarrow \frac{1}{4}\pi S^2 < S^2.$$

$$\text{Area of a square} = S^2, \text{since } d = S$$

Answer: B

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51. Let y be a real number.

Column A	Column B
$(y+1)\times y(y-1)$	$y\times y\times y$

A. The quantity in column A is greater.
B. The quantity in column B is greater.
C. The two quantities are equal.
D. The relationship cannot be determined.

Solution:

Column A: $(y+1)\times y\times(y-1)=(y^2+y)\times(y-1)$

$=y^3-y^2+y^2-y=y^3-y$

Column B: $y\times y\times y=y^3$

$\therefore\ y^3-y<y^3$.

Answer: B

DIRECTIONS: Questions **52-56** are based on a data given in the table below. Answer each of the these questions based on the given date

Year	Heavy	Light Commercial	Cars	Jeeps	Two-wheelers	Total
1996	21	32	202	146	349	750
1997	43	66	236	178	323	846
1998	74	75	232	217	402	1000
1999	86	94	289	256	418	1143
2000	103	119	261	235	476	1194
Total	327	386	1220	1032	1968	

Table: A data on the number of different categories of vehicles sold by a company over five years (in thousands).

52. In which of the following years was the number of light commercial vehicles sold was 25% of the number of two-wheelers sold on the same year?

A. 1996 B. 2000 C. 1998 D. 1999

Solution: 25% of T = L

25% of the 476 = $\frac{25}{100}\times 476=\frac{476}{4}=119$

Answer: B

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53. If the same percentage increase in the number of Heavy Vehicles as in 2000 over 1999 was expected in 2001 over 2000, approximately how many heavy vehicles were sold in 2001?

A. 138 B. 128 C. 123 D. 166

Solution: Percentage increase in 2000 over 1999 is

$$\frac{103-86}{86}\times 100\% = \frac{17}{86}\times 100\% = 19.76\%$$

Total number of Heavy Vehicles sold in 2001 is 103 + 19.76% of 103
$= 103 + 20.35 = 123.35 \approx 123$

Answer: C

54. Which one of the following is the percentage of the number of heavy vehicles sold in 1999 to that of the total of all vehicles sold in 1998?

A. 12% B. 7.5% C. 8.6% D. 9%

Solution: Number of Heavy vehicles sold in 1999 = 86

Total number of all vehicles sold in 1998 = 1000

$$\text{Percentage} = \frac{86}{1000}\times 100\% = 8.6\%$$

Answer: C

55. In which of the following years was the percentage of number two -wheelers sold to the total number of vehicles sold during the same year was highest?

A. 1999 B. 1996 C. 1998 D. 1997

Solution: Percentage highest if the denominator is smaller than the numerator or nearly equal.

Thus, in 1996, $\frac{349}{750}\times 100\% = 46.5\%$

and in 1999, $\frac{418}{1143}\times 100\% = 36.6\%$

Answer: B

56. For which of the following category of vehicles that the percentage increase in the sale in 1997 over the previous year was maximum?

A. Cars C. Jeeps

B. Two-wheelers D. Light commercial

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Solution:

$$\text{Cars}: \frac{66-32}{32} \times 100\% = \frac{34}{32} \times 100\% = 106.25\%$$

Answer: D

57. A container has 235 flowers in a variety of colors: red, blue, yellow, and purple. There are 45 blue flowers; one-fifth of the flowers are red; forty percent of them are purple and the rest are yellow. A flower is picked at random from the container. Which one of the following is the probability that the flower picked is yellow?

A. $\frac{1}{5}$ B. $\frac{2}{5}$ C. $\frac{49}{235}$ D. $\frac{9}{47}$

Solution: Total number of flowers = 235

$$\text{Blue} = 45$$

$$\text{Red} = \frac{1}{5} \cdot 235 = 47$$

$$\text{Purple} = 40\% \times 235 = \frac{40}{100} \times 235 = 94$$

$$\text{Yellow} = 235 - (45 + 47 + 94) = 235 - 186 = 49$$

$$P(\text{yellow}) = \frac{49}{235}$$

Answer: C

58. You go to a restaurant for lunch and want to order a sandwich. There are twenty-five sandwiches on the menu and six of them are toasted. If you order a sandwich at random, then what is the probability of your ordering a sandwich is **NOT** toasted?

A. $\frac{6}{25}$ B. $\frac{9}{25}$ C. $\frac{16}{25}$ D. $\frac{19}{25}$

Solution: Number of sandwiches = 25

Number of toasted sandwiches = 6

$$P(\text{toasted sandwiches}) = \frac{6}{25}$$

$$P(\text{not toasted}) = 1 - \frac{6}{25} = \frac{25-6}{25} = \frac{19}{25}$$

Answer: D

59. 28 male lions and 172 female lions are living in 500-acre Park. If 35

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more male lions are introduced into the park, how many more female lions should be introduced so that $\frac{6}{7}$ of the total number of lions in the Park is female?

A. 378 B. 206 C. 324 D. 241

Solution: Total Male $= 28 + 35 = 63$

Total Female $= 172 + F_{added}$

$$\frac{6}{7}(\text{Total}) = \text{Female total}$$

$$\text{Total} = \frac{7}{6}F_{total}$$

$$63 + (172 + F_{added}) = \frac{7}{6}(172 + F_{added})$$

$$6(235 + F_{added}) = 7(172 + 7F_{added})$$

$$1410 + 6F_{added} = 1204 + 7F_{added}$$

$$1410 - 1204 = F_{added}$$

$$206 = F_{added}$$

Answer: B

60. A business is owned by 1 man and 5 women, each of whom, has equal share. If one of the women sells $\frac{2}{5}$ of her share to the man, and another woman keeps $\frac{1}{4}$ of her share and sells the rest to the man, what fraction of the total share of the business is owned by the man?

A. $\frac{9}{40}$ B. $\frac{37}{120}$ C. $\frac{2}{3}$ D. $\frac{43}{120}$

Solution: Man's share $= \frac{1}{6}$ of the total share.

$$\text{Total Man's share} = \frac{1}{6} + \frac{2}{5}\left(\frac{1}{6}\right) + \frac{3}{4}\left(\frac{1}{6}\right)$$

$$= \frac{1}{6} + \frac{1}{15} + \frac{1}{8} = \frac{43}{120}$$

Answer: D

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SECTION ONE: Verbal Reasoning (Questions 1 - 35)

This section which is subdivided in to analogy, synonyms, antonyms classification, reading comprehension, analytical and logical reasoning contains 35 verbal reasoning items altogether. Each subsection is followed by its specific direction.

Reading Comprehension I (1 - 5)

Direction: Read the passage carefully and answer questions 1 – 5. Each question will be based on the information stated or implies in the passage. Then **blacken** the letter of your choice on the separate answer sheet provided.

If a person suddenly encounters any terrible danger, the change of nature one undergoes is equally great. Sometimes fear numbs our senses. **Like animals, one** stands still, powerless to move a step in fright or to lift a hand in defense of our lives, and sometimes one is seized with panic, and again, act more like the inferior animals than rational beings. On the other hand , frequently in cases of sudden extreme peril, which cannot be escaped by flight, and must be instantly faced, even the most timid men at once as if by miracle, become possessed of the necessary courage, sharp quick apprehension, and swift decision. This is a miracle very common in nature. Man and the inferior animals alike, when confronted with almost certain death 'gather resolution from despair' but there can really be no trace of so **debilitating** a feeling in the person fighting, or prepared to fight for dear life. At such times the mind is clearer than it has ever been; the nerves are steel, there is nothing felt but a wonderful strength and daring. Looking back at certain perilous moments in my own life, I remember them with a king of joy, not that there was any joyful excitement then but because they broadened my horizon, lifted me for a time above myself. Adapted from Reading Comprehension - Practice Questions

1. The title that best the passage would be.
   A. The Change of Nature
   B. Courage and panic
   C. The Will to Fight
   D. The Miracle of Confronting Danger

Answer: B

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2. What is the meaning of the world debilitating?

A. Strengthen
C. Enfeeble
B. Fatten
D. Debase

Clue: debilitating : something causing a loss of energy / strength (weakening) **enfeeble** = to make feeble or weaken
= to become less strong

Answer: C

3. The author feels happy in the recollection of dangers faced and over-come because

A. these experiences boosted his confidence.
B. he felt elated as he was alive
C. they brought him a new experience.
D. they added lifted him above himself for a time.

Answer: D

4. A man may react to sudden danger in some different ways. Some of these are:

A. He may be paralysed with fear, seized with panic, or act like an inferior animal.
B. He may be paralysed with fear, run a way, or fight.
C. He may flee panic, or fight back, stand still or act like an inferior animal.
D. He may be paralysed with fear, flee panic, seized with panic or act like an inferior animal.

Answer: A

5. Explain the phrase 'gather resolution from danger'.

A. To remain calm and not to lose hope when there is a difficulty
B. Questions to be enthusiastic and brave the odds during difficulty.
C. Hopelessness makes one determined to face the difficulty
D. Find peace in times of difficulty.

Answer: A

Classification Items (6 - 10)

Directions: In questions 6 - 10 a set of four words is given. Three of the words are related in some ways. The remaining word is not related to the

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rest. Choose the word that does not fit the relation and blacken the letter of your choice on the separate answer sheet provided.

6. A. Leopard C. Lion
   B. Tiger D . Elephant

Clue: Leopard, tiger and lion are cat species, but an elephant is not.

Answer: D

7. A. Pen B. Eraser C. Book D. Pencil

Clue: Pen, pencil and eraser are stationery materials but book is not.

Answer: C

8. A. Legal B. Order C. Chaos D. Confusion

Clue: Legal, order and chaos are somewhat related words (they are of law or rules) but confusion is not (it means vague, unclear)

Answer: D

9. A. Legal B. Tangue C. Eye D. Heart

Clue: Tangue, ear and eye are sense organs but heart is not.

Answer: D

10. A. Publisher B. Reporter C. Writer D. Reader

Clue: writer, publisher and reader are closely related (what written is published, what published is read) but reporter is related to news.

Answer: B

Reading Comprehension II (11 - 15)

Directions: Read the passage carefully and answer questions **11 - 15.** Each question will be based on the information stated or implied in the passage. Then **blacken** the letter of your choice on the separate answer sheet provided.

Daylilies are beautiful perennial flowers that can brighten up any yard or landscape. They are tolerant of drought and flooding, immune to heat stress, and grow well in full sun or light shade. They are the perfect choice for just about any soil or climate condition.

Different varieties of daylilies can be in bloom from late spring until autumn. Individual flowers last only one day, but each plant produces many buds, and many varieties have more than one flowering period.

Daylilies grow best in soil that is slightly acidic, and they prefer either direct sunshine or light shade. The best time to plant them is in the early

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fall or early spring, but they are hardy enough to endure planting or transplanting at almost any time of year. They should be planted 18 to 24 inches apart, and the bulb should be no deeper than one inch below the soil's surface. Adapted from **Reading Comprehension - practice Questions**

11. According to the passage, what soil is best for daylilies?

A. Any soil
C. Slightly neutral
B. Moist soil
D. Slightly acidic

Answer: D

12. You could infer from this passage that these flowers are called day - lilies because

A. The blooms last for one day.
B. Charles Day developed the hybrid.
C. They only bloom in the daytime.
D. They look like real lilies, but aren't

Clue: The daylilies are perennial plants that have flessy roots grass like leaves and colourful flowers that bloom for just one day.

Answer: A

13. How deep should daylilies be planted, according to the passage?

A. 18 to 24 inches
C. one to 20 inches
B. any depth work
D. one inch or more

Answer: A

14. Daylilies grow best in

A. the evenings
C. darkness
B. sunshine
D. the mornings

Clue: They grow well in full sun or light shade.

Answer: B

15. Which of the following is true according to the passage?

A. Heat can cause stress to daylilies.
B. Daylilies can not grow in sunny weather.
C. Daylilies can stand drought
D. Daylilies cannot stand drought.

Answer: C

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Antonyms (16 - 18)

Directions: Question **16 - 18** consist of sentences in which one word is written in bold followed by four other words. Choose the word that is most **nearly opposite in** meaning to the word in **bold.** Then, blacken the letter of your choice on the separate answer sheet provided.

16. The officer exaggerated the damage caused by the rowdies and thus made it look more serious.

A. Unreported
B. Condemned
C. Ignored
D. Underestimated

Clue: exaggerated = overstated, hyperbolized,made bigger up
exaggerated ≠ underestimated, lowered

Answer: D

17. The speaker was unable to **pacify** the crowd.

A. Antagonize
B. Challenge
C. Excite
D. Threaten

Answer: A

18. His negative attitude to poor people is deplorable.

A. Equitable
B. Desirable
C. Miserable
D. Commendable

Clue: deplorable = very bad and unacceptable with people
equitable = fair / good and reasonable with people.

Answer: A

Analytical and Logical Reasoning Questions (19 - 26)

Directions: Questions **19 - 26** deal with logical and analytical reasoning. Read the given set of conditions and assumption for each question. Choose the answer that you think is most appropriations among the given alternatives. Then **blacken** the letter of your choice on the separate answer sheet provided.

19. Tamrat's train never bothered him because there were always seats available on it, and he was able to spend his time on the train comfortably reading the newspaper or playing games on his mobile phone. Ever since the train schedule changed, train has been extremely crowded, and by the time the doors open at his station, there is not a seat to be found.

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A. Any traveller who uses the train will complain about the new schedule

B. Tamrat will likely look for anew job closer to home.

C. Tamrat would be better off taking off taking the bus to work instead of the train

D. Tamrat's train is less comfortable since the train schedule changed

Answer: D

20. It is called Restitution when you compensate someone for damaging his or her property in some way, which of the following situations is the best example of Restitution?

A. Peter asks tom to stay in his apartment when he is out of town. One day Tom arrives to stay and finds that pipe has burst and the apartment is filled with water. He calls the plumber to repair the pipe and pays for the repair

B. A pothole in the parking of Peter's company caused his flat tyre. He informs his boss and expects that the company should pay for the repair.

C. Tom borrows his friend's car and returns the car with an empty petrol tank. He apologizes and tells his friend that he will fill the tank tomorrow.

D. Tom borrows his friend's camera. Tom fails to zipper the case, and the camera falls on the ground, and the lens shatters. When Tom returns the camera, he tells his friend that he will pay for the damage.

Clue: The definition is: the act or the payment, usually, money for some harm or wrong that someone has suffered.

Answer: D

21. A four - person crew is painting Mr. Odom's house. Michael is painting the front of the house. Roza is in the alley behind the house painting the back. Jemal is painting the window frames on the north side, shawul is on the south. If Michael switches places with Jamal, and Jemal then switches places with shawul, where is shawul?

A. in front of the house

B. on the south side of the house

C. in the alley behind the house

D. on the north side of the house

Answer: D

22. Four friends in the sixth grade were sharing a pizza. They decided that

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the oldest friend would get the extra piece. Hawi is two months older than Geremew, who is three months younger than Nardos. Kemal is one month older than Germew. who should get the extra piece of Pizza?

A. Geremew B. kemal C. Hawi D. Nardos

Answer: D

23. If you are a fitness walker, you don't need to go to a health gym. you also don't need any fitness equipment for workouts. All you need is a pair of comfortable athletic shoes. Which of the following statements does this paragraph?

A. Fitness walking is an effective and convenient form of exerc

B. poorly designed shoes can cause foot injuries

C. Fitness walking is better than weight lifting

D. Walking outdoors provides more health benefits than walking indoors.

Answer: D

24. Sara lives in Adama. Her younger cousin Martha lives in a small town in the eastern part of Ethiopia. Martha has visited Sara Several times during the past five Years. In the same period of time, Sara has visited Martha only once

A. Sara is older than Martha

B. Martha wants to move to another town

C. Martha likes Sara better than Sara likes Martha.

D. Sara thinks small towns are boring.

Answer: D

25. Solomon introduced Sara as the mother of his sister's brother. How is Sara related to solomon?

A. Mother B. Sister C. Aunt D. Niece

Answer: A

26. **Mrs**. Genet wishes to renovate her cottage. She hires the services of a plumber, a carpenter, a painter, an electrician, and an interior decorator. The renovation is to be completed in a period of one week, i.e. Monday to Friday. Every worker will be taking one complete day to do this job. Mrs. Genet will allow just one person to work per day.

➥ The painter can do this work only after the plumber and the car-

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penter have completed their jobs.

➥ The interior decorator has to complete his job before that of the electrician.

➥ The carpenter cannot works on Monday or Tuesday.

In case the painter works on Thursday, which among the following alternative is possible?

A. The interior decorator does his work after the painter

B. The plumber and the painter work on consecutive days.

C. The electrician works on Tuesday

D. The electrician works on Friday.

Answer: D

Analogy (27 - 31)

Directions: To answer the items **27 - 31** read each of the following words carefully and choose the word that correctly completes the analogy. Then blacken the letter of your choice on the separate answer sheet provided.

27. EXERCISE is to GYM as EATING is to: _____

A. Drink B. Restaurant C. Food D. Dieting

Clue: gym is a fitness centre, health club, (sport) (where different kinds of exercise done) **Restaurant** is eating centre (at) in which food and drinks are served.

Answer: B

28. The EVENING is to MORNING as DINNER is to: _____

A. Lunch B. Brunch C. Supper D. Breakfast

Clue: Dinner is served in the evening and **breakfast** is served in the morning (opposite analogies)

Answer: D

29. GUN is to BULLET as a BOW is to: _____

A. Fly B. Sky C. Arrow D. Wood

Clue: Gun is used for shooting bullets .

Bow is used for shooting arrows.

Answer: C

30. PEN is POET as NEEDLE is to _____

A. Sewing B. Tailor C. Thread D. Button

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Clue: A **poet** uses a **pen** to write poems.

A **tailor** use a **needle** to sew clothes

31. PLAY is to ACTOR as CONCERT is to:_____ **Answer: B**

A. piano
C. symphony
B. persuasion
D. musician

Clue: An **actor** takes part in or acts in a theatrical play, in a film, in drama, etc.

A **musician** takes part in or performs a concert to show the performance.

Answer: D

Synonyms (32 - 35)

Directions: Questions 32 - 35 consist of sentences in which one word is written in bold followed by four other words. Choose the word that is **most nearly similar** in meaning to the word in **bold**. Then, **blacken** the letter of your choice on the separate answer sheet provided.

32. The special TV report **disrupted** regular programming due to the day's special events.

A. Interrupted
C. Satisfied
B. Continued
D. Complemented

Answer: A

33. He **dissipated** his inheritance in less than six months and he is now back to where he was

A. Saved
B. Hoarded
C. Wasted
D. Support

Answer: C

34. If enough members of parliament **dissent**, the new law will not be approved.

A. Question
B. Disagree
C. Agree
D. Support

Answer: B

35. During the riot, some of the demonstrators **looted** shop and supermarkets and took the goods to their home.

A. burned
B. shot
C. robbed
D. wasted

Answer: C

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SECTION TWO: Quantitative Reasoning (Questions 36 - 60) below and select the best answer from the given alternatives and **blacken** the letter of your choice on the separate answer sheet.

Question **36 - 40** are based on the table below, which gives the expenditure of a firm (in millions of birr) per annum over the given years.

Year	Item of expenditure				
	Salary	Fuel and Transport	Bonus	Interest on Loans	Taxes
2006	314	92	3	24	85
2007	258	110	2.5	33	71
2008	732	131	3.4	28	112
2009	450	104	2.6	31	98
2010	370	125	3.8	42	93

36. The total expenditure of the firm over these items during the year 2007 is

A. 467.5 B. 476.5 C. 447.5 D. 474.5

Solution:

Total expenditure in 2007 $= 258 + 110 + 2.5 + 33 + 71 = 474.5$

Answer: D

37. What is the average amount of tax per year which the firm had to pay during the whole period?

A. 91.8 B. 82.9 C. 114.75 D. 76.5

Solution:

$$\text{Average} = \frac{85+71+112+98+93}{5} = \frac{459}{5} = 91.8$$

Answer: A

38. Total expenditure on all these items in the year 2006 was approximately what percent of the total expenditure in the year 2010?

A. 83.71 B. 81.73 C. 78.25 D. 80.12

Solution: Total expenditure in $2006 = 314 + 92 + 3 + 24 + 85 = 518$

Total expenditure in $2010 = 370 + 125 + 3.8 + 42 + 93 = 633.8$

$$\text{Percentage} = \frac{518}{633.8} \times 100\% = 81.73$$

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Answer: B

39. The total amount of bonus paid by the firm during the whole period is approximately what percent of the total amount of salary paid during this period?

A. 0.62 B. 0.58 C. 0.72 D. 0.68

Solution: Total amount of bonus $= 3 + 2.5 + 3.4 + 2.6 + 3.8 = 15.3$

Total amount of salary $= 314 + 258 + 732 + 450 + 370 = 2124$

$$\text{Percentage} = \frac{15.3}{2124} \times 100\% = 0.72$$

Answer: C

40. What is the approximate ratio between the total expenditure on interest on loans for all the years and total expenditure on fuel and transport for all the years?

A. 10:36 B. 10:39 C. 10:26 D. 10:29

Solution:

Total expenditure on interest $= 24 + 33 + 28 + 31 + 42 = 158$

Total expenditure on fuel & transport $= 92 + 110 + 131 + 104 + 125 = 562$

$I : t = 158 : 562 = 79 : 281 \approx 10 : 36$ Answer: A

41. Let $x \neq -2, x \neq 3$ and $x \neq \frac{4}{7}$ then $\left(\frac{2}{x+2} + \frac{5}{x-3}\right) \div \left(\frac{7x+4}{2x-6}\right)$ equals :

A. $\frac{3}{x-3}$ B. $\frac{7x+4}{x+2}$ C. $\frac{2}{x+2}$ D. $\frac{7x-1}{(x-2)(7x+4)}$

Solution:

$$\left(\frac{2}{x+2} + \frac{5}{x-3}\right) \div \left(\frac{7x+4}{2x-6}\right)$$

$$\frac{2(x-3)+5(x+2)}{(x+2)(x-3)} \times \frac{2x-6}{7x+4}$$

$$\frac{7x+4}{(x+2)(x-3)} \times \frac{2(x-3)}{7x+4} = \frac{2}{x+2}$$

Answer: C

42. When the polynomial $5x^4 - 2x^3 + x^2 - x + 1$ is divided by $x^2 - 1$, the remainder is :

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A. $5x^2-2x+6$ B. $\frac{x^2}{-3x+7}$ C. $\frac{-3x+7}{x^2-1}$ D. $-3x+7$

$$x^2-1\overline{)\,5x^4-2x^3+x^2-x+1}$$

$5x^2-2x+6 \leftarrow$ quotient

$5x^4-5x^2$

$-2x^3+6x^2-x+1$

$-2x^3+2x$

$6x^2-3x+1$

$6x^2-6$

$-3x+7 \leftarrow$ remainder

Answer: D

43. Given that x and y are negative numbers

Quantity A	Quantity B
$x+y$	$x-y$

Which of the following is true about the relationship between quantities A and B ?

A. Quantity A is greater than quantity B.

B. Quantity B is greater than quantity A.

C. The two quantities are equal.

D. The relationship cannot be determined.

Solution:

For all negative real number, x and y, $x-y>x+y$

Answer: B

44. Given that $2^{1.5894}=3$

Quantity A	Quantity B
$2^{4.327}$	11

How are quantities A and B related?

A. Quantity A is greater than quantity B.

B. Quantity B is greater than quantity A.

C. The two quantities are equal.

D. The relationship cannot be determined.

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Solution:

Quantity A: $2^{4.327} > 2^4 = 16$

Quantity B = 11

Quantity A > Quantity B

45. Two six side dice are thrown together. What is the probability that the sun of two numbers on top is 10?

Answer: A

A. $\frac{20}{81}$ B. $\frac{25}{81}$ C. $\frac{8}{81}$ D. $\frac{16}{81}$

Solution:

$n(S) = 6 \times 6$, $E = \{(5,5), (4,6), (6,4)\}$

$$P(E) = \frac{n(E)}{n(S)} = \frac{3}{36} = \frac{1}{12}$$

Answer: B

46. Nine identical balls numbered 1 trough 9 are placed in a bag A ball is randomly drawn and replaced. Then, second ball is randomly drawn. What is the probability that the first ball drawn and the second ball drawn are both even numbered?

A. $\frac{20}{81}$ B. $\frac{25}{81}$ C. $\frac{8}{81}$ D. $\frac{16}{81}$

Solution:

$n(S) = 9$, $E_1 = \{2,4,6,8\}$, $E_2 = \{2,4,6,8\}$

$$P(E_1 \text{ and } E_2) = P(E_1) \times P(E_2) = \frac{4}{9} \times \frac{4}{9} = \frac{16}{81}$$

Answer: D

47. Mesfin bought a laptop valued Birr 7700. One year later the value of the laptop is decreased by $\frac{2}{7}$. What is the new value of the laptop?

A. 5500 birr B. 9900 birr C. 2200 birr D. 4400 birr

Solution:

$$\text{The new value of the laptop} = \text{Birr } 7700 - \frac{2}{7} \times \text{Birr } 7700$$

$$= \text{Birr } 5500$$

Answer: A

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48. Hamza spent 22 minutes on his English homework and 25.08 minutes on his Mathematics homework. What percent more time does Hamza spent on his Mathematics homework?

A. 87 B. 28 C. 14 D. 12

Solution:

The additional time spent on Mathematics = 25.08 – 22 = 3.08minute

$$\text{Percentage} = \frac{3\cdot 08}{22} \times 100\% = 14\%$$

Answer: C

49. Meseret begins a saving program in which she will save Birr 1000 the first year, and each subsequent year will save Birr 200 more than the she did the previous year. How much will she save during the eighth year?

A. Birr 2600 B. Birr 1360 C. Birr 2200 D. Birr 2400

Solution:

1000, 1200, 1400, ... are arthimetic progration

$A_7 = A_1 + (7-1)d$ --- (during the 8th year = A_7)

$= 1000 + 6 \times 200 = 200$ Birr

Answer: C

50. A ball bounces one-half the distance from which it falls. How high does it bounce up on the fifth rebound if the ball is dropped from a height of 20m?

A. 0.625m B. 1.25m C. 5m D. 2.5m

Solution:

Since the model from a geometric progression,

$h(n) = ab^n$ where h(n) = height on bounce n

a = starting height = 20m

b = common ratio = $\frac{1}{2}$

n = number of bounces = 5

$$h(5) = 20 \cdot \left(\frac{1}{2}\right)^5 = \frac{20}{32} = 0\cdot 625\text{m}$$

Answer: A

51. A contest offers a total of 18 prizes. The first prize is worth Birr 10,000, and each successive prize is worth Birr 500 less than the next higher prize. What is the total value of all the prizes?

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A. Birr 108,000
B. Birr 105,000
C. Birr 103,500
D. Birr 113,500

Solution:

$$10,000, 9500, 9000, \cdots \longleftarrow \text{A.P}$$

$$S_n = \frac{n}{2}[2A_1 + (n-1)d]$$

$$S_{18} = \frac{18}{2}[2\times 10,000 + 17\times(-500)] = 103,500\text{Birr}$$

Answer: C

52. A student earns 13 birr per hour working for the computer center's help desk and 9.50 birr per hour working at the copy center. Because of her course load, she limits her work to 25 hours per week. She wants to earn at least 275 birr each week. Which of the following system of linear inequalities represent this information?

A. $\begin{cases} 13x + 9\cdot 5y < 275 \\ x + y \le 25 \\ x \ge 0, y \ge 0 \end{cases}$

B. $\begin{cases} 13x + 9\cdot 5y > 275 \\ x + y \le 25 \\ x \ge 0, y \ge 0 \end{cases}$

C. $\begin{cases} 13x + 9\cdot 5y \le 275 \\ x + y \ge 25 \\ x \ge 0, y \ge 0 \end{cases}$

D. $\begin{cases} 13x + 9\cdot 5y \ge 275 \\ x + y \le 25 \\ x \ge 0, y \ge 0 \end{cases}$

Solution:

Let x = number of working hours at the computer center per a day

y = number of hours at the copy center day

Then $x \ge 0, y \ge 0$

$x + y \le 25$

$13x + 9.5y \ge 275$

Answer: D

53. Consider the following system of linear inequalities

$$x - y \le 3$$
$$3x - 3y \ge d$$

If the solution set of the system is empty, which of the following numbers could be the value of d?

A. 8
B. 15
C. 5
D. 7

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Solution:

$x-y\leq 3$ and $3x-3y\geq d$

$y\geq x-3$ and $y\leq x-\frac{1}{3}d$

The solutions set of the system

$$\begin{cases} y\geq x-3 \\ y\leq x-\frac{1}{3}d \end{cases}$$ is empty if and only if $-3>-\frac{1}{3}d$

$d>9$

(∵ The region must have number common element)

Answer: B

54. A Pediatrician uses the model $h = 3.3a + 28.6$ to estimate the height h of a boy, in inches, in terms of the boy's age a, in years, between the age of 2 and 5. Based on the model, what is the estimated increase, the inches, of the boy's height each year?

A. 3.3 B. 4.3 C. 14.3 D. 28.6

Solution:

$h_a = 3.3a + 28.6$ and $h_{a+1} = 3\cdot 3(a+1) + 28.6$

$\Delta h = h_{a+1} - h_a = 3\cdot 3a + 3\cdot 3 + 286 - (3\cdot 3a + 28\cdot 6) = 3\cdot 3$

Answer: A

55. Suppose a chain letter is started by sending a letter to five people and the letter requires each person who receives the letter to send it on to five additional people. Assume that no person receives more than one letter. How many letters are sent all together if the chain reaches the eighth stage?

A. 490,625 B. 390,625 C. 488,280 D. 480,280

Solution:

$5, 5^2, 5^3, ---, 5^7, 5^8$

The letters sent all together $= 5^8 = 390{,}625$

Answer: B

56. If the line passing through (x, y) and $(-1, 2)$ has a slope of 3 and the line passing through (x, y) and $(2, -1)$ has a slope of 1, then (x, y) equals:

A. (4, 7) B. (−4, −7) C. (−4, 7) D. (4, −7)

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Solution:

$$m_1 = \frac{2-y}{-1-x} \quad , \quad m_2 = \frac{-1-y}{2-x}$$

$$3 = \frac{2-y}{-1-x} \qquad 1 = \frac{-1-y}{2-x}$$

$$3x - y = 5 \qquad x - y = 3$$

$$\begin{cases} 3x - y = -5 \\ x - y = 3 \end{cases}$$

$$x = -4,\ y = -7$$

Answer: B

57. Ali wants to invest a total of Birr 20,000 so that his yearly interest is Birr 1690. If the invests part of it in a bank yielding 8% profit and the other part in a stock at 9.5% profit, how much should he invest at each rate?

A. Birr 7500 at 9.5%; Birr 12500 at 8%

B. Birr 9000 at 9.5%; Birr 11000 at 8%

C. Birr 6000 at 9.5%; Birr 14000 at 8%

D. BIrr 8000 at 9.5%; Birr 12000 at 8%

Solution:

Let x = Amount of Birr invested in a bank

Then $20,000 - x$ is the amount of Birr invested in a stock since

$I = PRT$

$1690 = x\,8\% + (20,000 - x)\,9.5\%$

$$1690 = \frac{8}{100}x + \frac{190,000 - 9 \cdot 5x}{100} = -1 \cdot 5x = -21000$$

$x = 14,000$ and $20,000 - 14,000 = 6000$

Answer: C

58. The red, green and black lollies in a packet are in the ratio 2:3:5. How many green and black lollies are there, if there are 10 red lollies?

A. 25 green lollies and 15 black lollies

B. 30 green lollies and 50 black lollies

C. 6 green lollies and 15 black lollies

D. 15 green lollies and 25 black lollies

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Solution:

$R:G:B = 2:3:5$

$R:G = 2:3$ and $G:B = 3:5$

$\frac{R}{G} = \frac{2}{3}$ and $\frac{G}{B} = \frac{3}{5}$

$\frac{10}{G} = \frac{2}{3}$ $\frac{15}{B} = \frac{3}{5}$

$G = 15$ $B = 25$

Answer: D

59. At a restaurant the bill came to 82 birr. If you leave 119.72 birr, what percent tip is that?

A. 31.5 B. 46 C. 64 D. 38

Solution:

$$\text{Tip} = 119 \cdot 72\text{Birr} - 82\text{Birr} = 37 \cdot 72\text{Birr}$$

$$\text{Percentage} = \frac{37 \cdot 72}{82} \times 100\% = 46\%$$

Answer: B

60. Suppose the population of a town is given by the model

$p(t) = 17250\left(\frac{1}{2}\right)^{\frac{t}{10}}$, where t denotes the number of years since 2000.

Which of the following statement is true?

A. The population is halved every ten years.

B. The population is growing by 50% every ten years.

C. The population in 2000 was 8625.

D. The population doubles every ten years.

Solution:

$$P(t) = 17,250\left(\frac{1}{2}\right)^{\frac{t}{10}}$$

$$\text{when } t = 10,\ P(10) = 17,250\left(\frac{1}{2}\right)^{\frac{10}{10}} = \frac{1}{2}(17,250)$$

$\therefore$ The population is halved every 10 years. **Answer: A**

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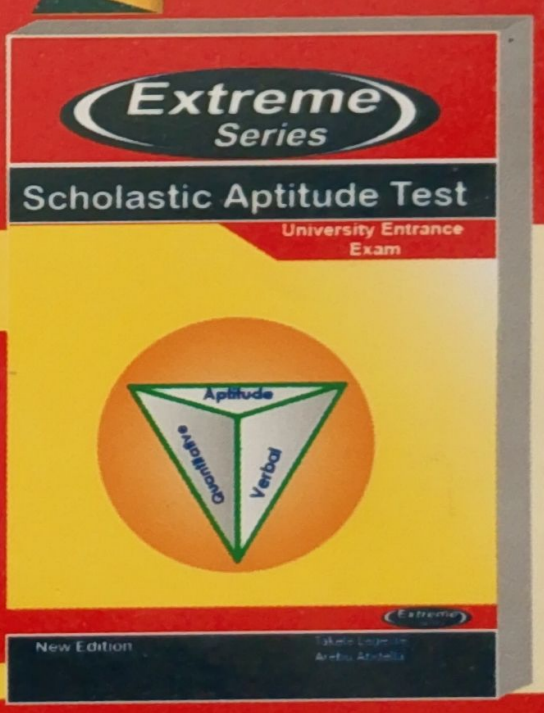

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